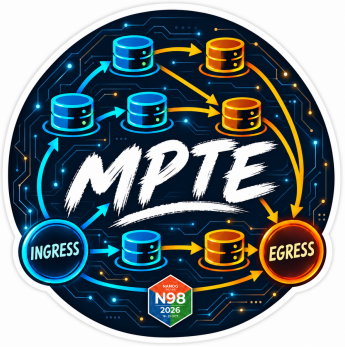
# Directed Acyclic Graph (DAG) based Multipath Traffic Engineering (MPTE)

## Workshop Activity Sheet

*Provisioning Multipath Traffic Engineering Directed Acyclic Graph (MPTED) Tunnels using RSVP over an MPLS Forwarding Plane with Distributed Label Assignment*

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## The lab at a glance

| Item | Value |
|---|---|
| Routers | R1 to R10, Junos cRPD containers in a containerlab topology |
| Router IDs | 10.123.123.1 (R1) to 10.123.123.10 (R10) |
| Links | 27 point-to-point links, IS-IS level 1 and 2, all with metric 10 and 1000 Mbps |
| Tunnel ingress | R1 (all MPTE tunnels start at R1 and end at R2, R8, R9 or R10) |
| Starting state | 30 MPTE tunnels are configured on R1 and all of them are deactivated |
| Reference | MPTE_NANOG_Cheatsheet.txt (all commands used here, plus more) |

## How to use this sheet

Every block you can copy is shown in a monospace font. Copy one block at a time.

```
this is a command: copy it into the router CLI
```

```
this is sample output: you should see something like this
```

**PROMPT (copy into Copilot chat)**

this is a prompt: copy it into Copilot chat with the Junos MCP server enabled

**A good answer includes:** what a correct answer to the prompt contains (your wording will differ)

**Config change:** steps that change the router configuration are marked like this and say how to undo them.

## Connect to R1

Your instructor will provide the details for connecting to R1 and the other routers. Once connected you get the operational prompt (root@R1>). Type configure to enter configuration mode, commit to apply changes and exit to leave a mode.

# Section 1. Understand the topology

Before bringing up any tunnel, look at the network the tunnels will run over. MPTE computes its paths from the Traffic Engineering Database (TED), so reading the TED is the first skill.

## 1.1 The lab topology

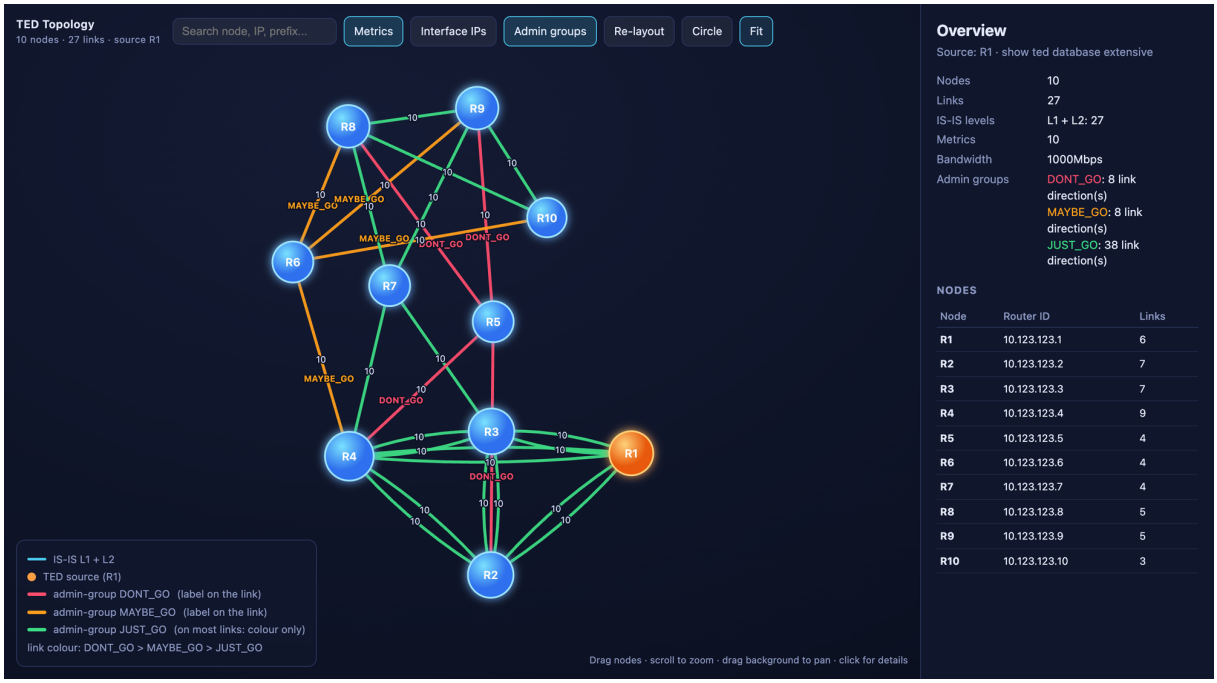


*Figure 1. The lab topology as generated from the TED of R1. R1 (orange) is the source of the TED; link colors show the admin-group; the number on each link is its metric.*

### Nodes

There are 10 nodes, R1 to R10. Each node is a router; its router ID (the loopback address) identifies it in the TED and in MPTE, where a router in a tunnel's DAG is called a junction.

| Node | Router ID | Links |
|---|---|---|
| R1 | 10.123.123.1 | 6 |
| R2 | 10.123.123.2 | 7 |
| R3 | 10.123.123.3 | 7 |
| R4 | 10.123.123.4 | 9 |
| R5 | 10.123.123.5 | 4 |
| R6 | 10.123.123.6 | 4 |
| R7 | 10.123.123.7 | 4 |
| R8 | 10.123.123.8 | 5 |
| R9 | 10.123.123.9 | 5 |
| R10 | 10.123.123.10 | 3 |

### Links

- There are 27 links. A link is seen from both ends, so the TED holds 54 link directions and the CLI shows 54 rows.

- R1, R2, R3 and R4 are joined by parallel links: every pair among them has two links, drawn as separate lines.
- The links toward R5 to R10 form the rest of the network. R5 and R6 sit between the R1 to R4 group and R7, R8, R9 and R10.
- The color of a link shows its admin-group (see below). Most links are green.

## 1.2 Key link attributes

Each link direction in the TED carries the attributes that tunnels use. This is the first link of R1, toward R2:

| Attribute | Example (R1 to R2) | What it means |
|---|---|---|
| Local / Remote address | 10.1.1.1 / 10.1.1.2 | Interface addresses at each end of the link |
| Interface index | local 632, remote 631 | Interface numbers at each end (they differ per run) |
| Metric (TE metric) | 10 | Cost used by MPTE to pick the shortest paths |
| IGP metric | 10 | Cost used by IS-IS routing |
| Static BW | 1000Mbps | Total bandwidth of the link |
| Reservable BW | 1000Mbps | Bandwidth tunnels may reserve on the link |
| Available BW [0..7] | 1000Mbps at each priority | Bandwidth still free, per priority; falls as tunnels reserve |
| Color (admin-group) | 0x8 JUST_GO | A tag on the link that tunnels can include or exclude |

### Admin-groups

Admin-groups are named tags that an operator puts on interfaces. Each name is a bit number; the color value in the TED is that bit as a mask. A tunnel can include or exclude links by their tag, which you will do in a later section.

| Admin-group | Bit | Mask | Links | Between |
|---|---|---|---|---|
| DONT_GO | 1 | 0x2 | 4 | R2-R5, R4-R5, R5-R8, R5-R9 |
| MAYBE_GO | 2 | 0x4 | 4 | R4-R6, R6-R8, R6-R9, R10-R6 |
| JUST_GO | 3 | 0x8 | 19 | all other links |

> **Worth noticing:** All 27 links have TE metric 10, IGP metric 10 and 1000 Mbps. Only the admin-group differs from link to link, so in this lab the admin-group, not the metric, is what makes tunnels choose different paths.

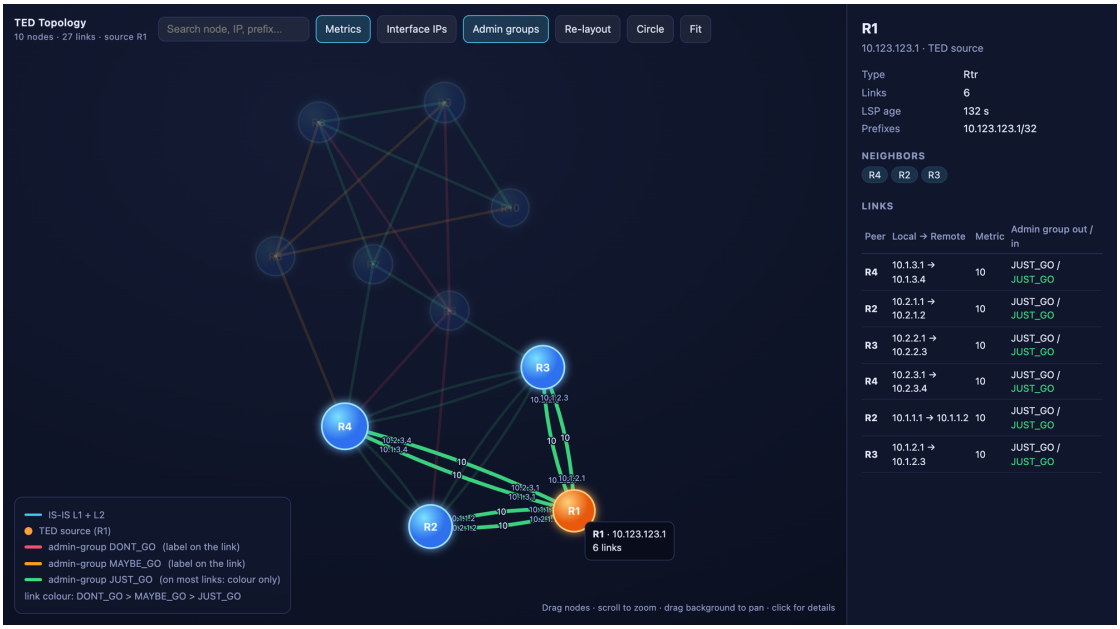


*Figure 2. Click a node to see its links. For R1 the panel lists each peer, the local and remote address, the metric and the admin-group in each direction.*

## 1.3 Show it from the CLI

### The nodes

```
show ted database extensive | match NodeID
```

```
NodeID: R1.00(10.123.123.1)
NodeID: R2.00(10.123.123.2)
NodeID: R3.00(10.123.123.3)
NodeID: R4.00(10.123.123.4)
NodeID: R5.00(10.123.123.5)
NodeID: R6.00(10.123.123.6)
NodeID: R7.00(10.123.123.7)
NodeID: R8.00(10.123.123.8)
NodeID: R9.00(10.123.123.9)
NodeID: R10.00(10.123.123.10)
```

### The links

One row per link direction. Parallel links appear as separate rows.

```
show ted link
```

```
ID                          ->ID                        LocalPath LocalBW
R1.00(10.123.123.1)           R4.00(10.123.123.4)               0 0bps
R1.00(10.123.123.1)           R2.00(10.123.123.2)               0 0bps
R1.00(10.123.123.1)           R3.00(10.123.123.3)               0 0bps
R1.00(10.123.123.1)           R4.00(10.123.123.4)               0 0bps
R1.00(10.123.123.1)           R2.00(10.123.123.2)               0 0bps
R1.00(10.123.123.1)           R3.00(10.123.123.3)               0 0bps
R2.00(10.123.123.2)           R1.00(10.123.123.1)               0 0bps
R2.00(10.123.123.2)           R4.00(10.123.123.4)               0 0bps
...
```

```
show ted link detail
```

```
R1.00(10.123.123.1)->R4.00(10.123.123.4), Local: 10.1.3.1, Remote: 10.1.3.4
  Local interface index: 530, Remote interface index: 531
  LocalPath: 0, Metric: 10, IGP metric: 10, StaticBW: 1000Mbps, AvailBW: 1000Mbps
      Color: 0x8 JUST_GO
  localBW [0] 0bps  [1] 0bps  [2] 0bps  [3] 0bps
  localBW [4] 0bps  [5] 0bps  [6] 0bps  [7] 0bps
  AvailBW [0] 1000Mbps  [1] 1000Mbps  [2] 1000Mbps  [3] 1000Mbps
  AvailBW [4] 1000Mbps  [5] 1000Mbps  [6] 1000Mbps  [7] 1000Mbps
...
```

**Why 54 rows:** Every link shows up twice in this output: once as seen from each of its two routers.

### The full TED

The complete database, with every attribute from the table above. It is long (about 2000 lines), so filter it.

```
show ted database extensive
```

```
TED database: 10 ISIS nodes 10 INET nodes 0 INET6 nodes
NodeID: R1.00(10.123.123.1)
  Type: Rtr, Age: 110 secs, LinkIn: 6, LinkOut: 6
  Protocol: IS-IS(1), MI: 0, MT: 0
   10.123.123.1
    To: R2.00(10.123.123.2), Local: 10.1.1.1, Remote: 10.1.1.2
      Local interface index: 632, Remote interface index: 631
      Color: 0x8 JUST_GO
      Metric: 10
      IGP metric: 10
      Static BW: 1000Mbps
      Reservable BW: 1000Mbps
      Available BW [priority] bps:
    ...
```

### The admin-groups in use

Run the Junos command inside the device shell and filter it with grep, then sort -u keeps each value once.

```
start shell command "cli -c 'show ted database extensive' | grep 'Color' | sort -u"
```

```
      Color: 0x2 DONT_GO
      Color: 0x4 MAYBE_GO
      Color: 0x8 JUST_GO
```

Replace sort -u with sort | uniq -c to count them. The counts are double the number of link directions, because each node lists its links once under IS-IS level 1 and once under level 2.

```
start shell command "cli -c 'show ted database extensive' | grep 'Color' | sort | uniq -c"
```

```
     16       Color: 0x2 DONT_GO
     16       Color: 0x4 MAYBE_GO
     76       Color: 0x8 JUST_GO
```

### The IS-IS neighbors of R1

```
show isis adjacency
```

```
Interface             System         L State        Hold (secs) SNPA
eth1                  R2             3  Up                   21
eth2                  R3             3  Up                   20
eth3                  R4             3  Up                   25
eth4                  R2             3  Up                   23
eth5                  R3             3  Up                   19
eth6                  R4             3  Up                   20
```

R1 has six adjacencies, which matches the six links of R1 in the TED: two each to R2, R3 and R4.

## 1.4 Ask the LLM

The Junos MCP server lets the LLM run these commands and read the result for you. Try these prompts and compare each answer with the CLI output above.

**CLI and prompts:** The prompts are typed into Copilot chat, which calls the Junos MCP server for you. The MCP server runs the same Junos commands you see in the CLI blocks.

**PROMPT (copy into Copilot chat)**

Using the Junos MCP server, read the TED from router r1 and give me a table with one row per router: router ID and number of links.

**A good answer includes:** 10 routers, R1 to R10, with router IDs 10.123.123.1 to 10.123.123.10 and link counts R1=6, R2=7, R3=7, R4=9, R5=4, R6=4, R7=4, R8=5, R9=5, R10=3.

**PROMPT (copy into Copilot chat)**

Which links in the topology have an admin-group other than JUST_GO? For each, give the two routers, their interface addresses and the group.

**A good answer includes:** 8 links: R2-R5 (DONT_GO); R4-R5 (DONT_GO); R5-R8 (DONT_GO); R5-R9 (DONT_GO); R4-R6 (MAYBE_GO); R6-R8 (MAYBE_GO); R6-R9 (MAYBE_GO); R10-R6 (MAYBE_GO). Each has the same group in both directions.

**PROMPT (copy into Copilot chat)**

How many links are there between R1 and R2, and what are their interface addresses?

**A good answer includes:** Two parallel links: 10.1.1.1 to 10.1.1.2, and 10.2.1.1 to 10.2.1.2.

**PROMPT (copy into Copilot chat)**

Summarize the traffic-engineering attributes of the links in this topology. Is there any link whose metric or bandwidth differs from the others?

**A good answer includes:** All links have TE metric 10, IGP metric 10 and 1000 Mbps; nothing differs except the admin-group (DONT_GO, MAYBE_GO or JUST_GO).

# Section 2. Bring up compute-only DAGs

A compute-only tunnel asks the router to compute the DAG, the set of paths from the ingress to the egress, without signaling anything. No RSVP sessions are set up, no labels are allocated and no bandwidth is reserved. It is the safest way to see which paths MPTE would use.

The lab has 30 tunnels configured on R1, all deactivated. Their names tell you what they do: -cmp is compute-only and -margin-0 is a metric margin of 0. The destination is in the name (R1toR9 ends at R9). Start by confirming that nothing is active:

```
show mpte tunnel terse
```

```
Total: 0
Up 0, Degraded 0, Down 0
```

## 2.1 Compute-only: R1toR9-cmp

Activate the tunnel. Its configuration is already in place: an ingress, an egress and compute-only.

```
configure
activate protocols mpte tunnel R1toR9-cmp
show | compare
commit
exit
```

```
[edit protocols mpte]
!     active: tunnel R1toR9-cmp { ... }
```

**Undo:** run deactivate protocols mpte tunnel R1toR9-cmp and commit.

Check that it came up:

```
show mpte tunnel
```

```
Originator       ID     State  Avail  JCTs Type       SigType TunnelName
10.123.123.1     100    Up     100%   10   MPLS-Sig   Compute R1toR9-cmp

Total: 1
Up 1, Degraded 0, Down 0
```

SigType Compute means the tunnel is computed only. The tunnel is Up and has 10 junctions. Now look at the DAG that was computed:

```
show mpte tunnel name R1toR9-cmp extensive
```

```
R1toR9-cmp, Originator: 10.123.123.1, ID: 100
  Up, Availability: 100%, Type: MPLS-Sig
  ComputeEngine: 10.123.123.1, Compute-only
  Ingresses (1):
    10.123.123.1, BW 0bps
  Egresses (1):
    10.123.123.9
  OptimizationObjective: TE, MetricMargin: Infinite
  Metric Min 30 Max 40
  Max-Flow Throughput: 4Gbps
  DAGVersion 1
    Junctions (10):
    ...
    2 Oct 18 17:54:45.297 Configuration Changed
    1 Oct 18 17:54:45.084 Tunnel Created
  Created: Sun Oct 18 17:54:45 2026
Total: 1
Up 1, Degraded 0, Down 0
```

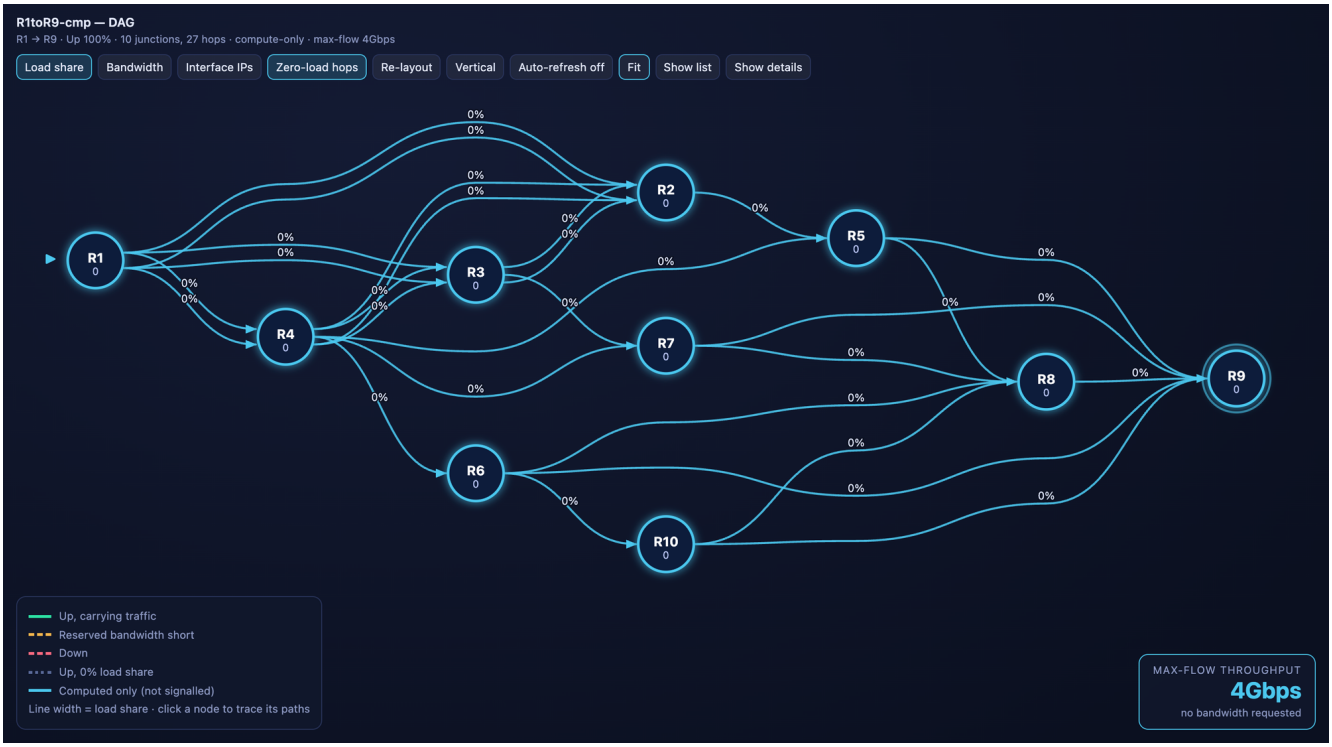


*Figure 3. DAG of R1toR9-cmp: every path MPTE could use from R1 to R9. Junctions are routers; edges are next hops.*

## How to read the picture

- **Nodes** are the junctions (routers). The triangle marks the ingress R1 and the ring marks the egress R9.
- **Edges** are next hops, one per interface. Parallel links between two routers are separate curves.
- **Cyan** means computed only. A signaled tunnel is drawn green.
- **0%** on each edge is the load share. With no bandwidth requested, no traffic is assigned to any hop.

**In numbers:** The DAG has 10 junctions and 27 hops, the metric from R1 to R9 ranges from 30 to 40, and the maximum throughput is 4Gbps. Max-flow is the most traffic the DAG can carry, limited by the 1000 Mbps links.

### Nothing is signaled

The compute-only tunnel installs no routes and sets up no RSVP sessions:

```
show route table inet.3
show rsvp mpte session
```

The first command prints nothing: there is no route to R9 through a tunnel. The second shows no sessions:

```
Local Ingress MPTE: 0 sessions, 0 up, 0 down, 0 degraded

Local Transit MPTE: 0 sessions, 0 up, 0 down, 0 degraded
```

### Ask the LLM

**Config change:** This prompt changes the router configuration. Undo: ask the LLM to deactivate the tunnel.

**PROMPT (copy into Copilot chat)**

Using the Junos MCP server on router r1, activate the MPTE tunnel R1toR9-cmp and show me its state. Is it signaled or compute-only?

**A good answer includes:** The tunnel is Up with 100% availability and SigType Compute (compute-only), with 10 junctions.

**PROMPT (copy into Copilot chat)**

On r1, summarize the extensive output of tunnel R1toR9-cmp: ingress, egress, number of junctions and hops, metric range and max-flow throughput. Why do all the junctions show Down?

**A good answer includes:** Ingress R1 (10.123.123.1), egress R9 (10.123.123.9), 10 junctions, 27 hops, metric 30 to 40, max-flow 4Gbps. The junctions show Down because a compute-only tunnel is never signaled, even though the tunnel itself is Up.

## 2.2 Compute-only with a bandwidth request

A tunnel can also request bandwidth. On a compute-only tunnel the request is used for the computation but never reserved. Add 400 Mbps to the ingress of R1toR9-cmp:

```
configure
set protocols mpte tunnel R1toR9-cmp ingress 10.123.123.1 bandwidth 400m
show | compare
commit
exit
```

```
[edit protocols mpte tunnel R1toR9-cmp ingress 10.123.123.1]
+     bandwidth 400m;
```

**Undo:** see below, where the bandwidth is removed again.

```
show mpte tunnel name R1toR9-cmp extensive
```

```
R1toR9-cmp, Originator: 10.123.123.1, ID: 100
  Up, Availability: 100%, Type: MPLS-Sig
  ComputeEngine: 10.123.123.1, Compute-only
  Ingresses (1):
    10.123.123.1, BW 400Mbps
  Egresses (1):
    10.123.123.9
  OptimizationObjective: TE, MetricMargin: Infinite
  Metric Min 30 Max 40
  Max-Flow Throughput: 4Gbps
  DAGVersion 2
    Junctions (10):
    ...
      JCT 10.123.123.1, Down, ReqBW 400Mbps
        NHOPs (6):
          NHOP 10.1.1.2 Index 631 Metric 30
            HopVersion 1, Down, OutLabel: 0
              LoadShare 12%, ReqBW 50Mbps, ResBW 0bps
          NHOP 10.2.1.2 Index 633 Metric 30
            HopVersion 1, Down, OutLabel: 0
              LoadShare 12%, ReqBW 50Mbps, ResBW 0bps
          NHOP 10.1.3.4 Index 531 Metric 30
            HopVersion 1, Down, OutLabel: 0
              LoadShare 25%, ReqBW 100Mbps, ResBW 0bps
          NHOP 10.2.3.4 Index 533 Metric 30
            HopVersion 1, Down, OutLabel: 0
              LoadShare 25%, ReqBW 100Mbps, ResBW 0bps
```

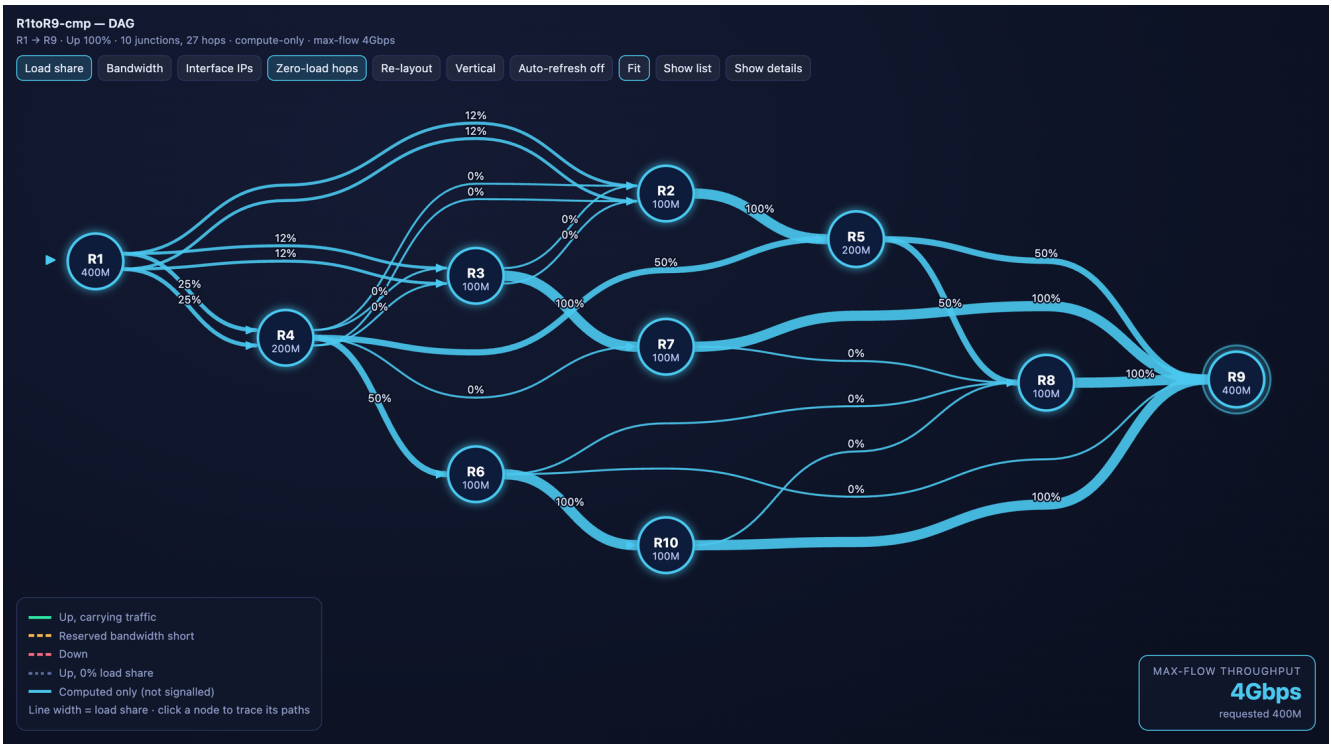


*Figure 4. DAG of R1toR9-cmp with 400 Mbps requested. The number inside each node is the bandwidth requested at that junction; the percentage on each edge is its share of the traffic.*

Turn on the Bandwidth button in the page to see the reserved bandwidth on every hop:

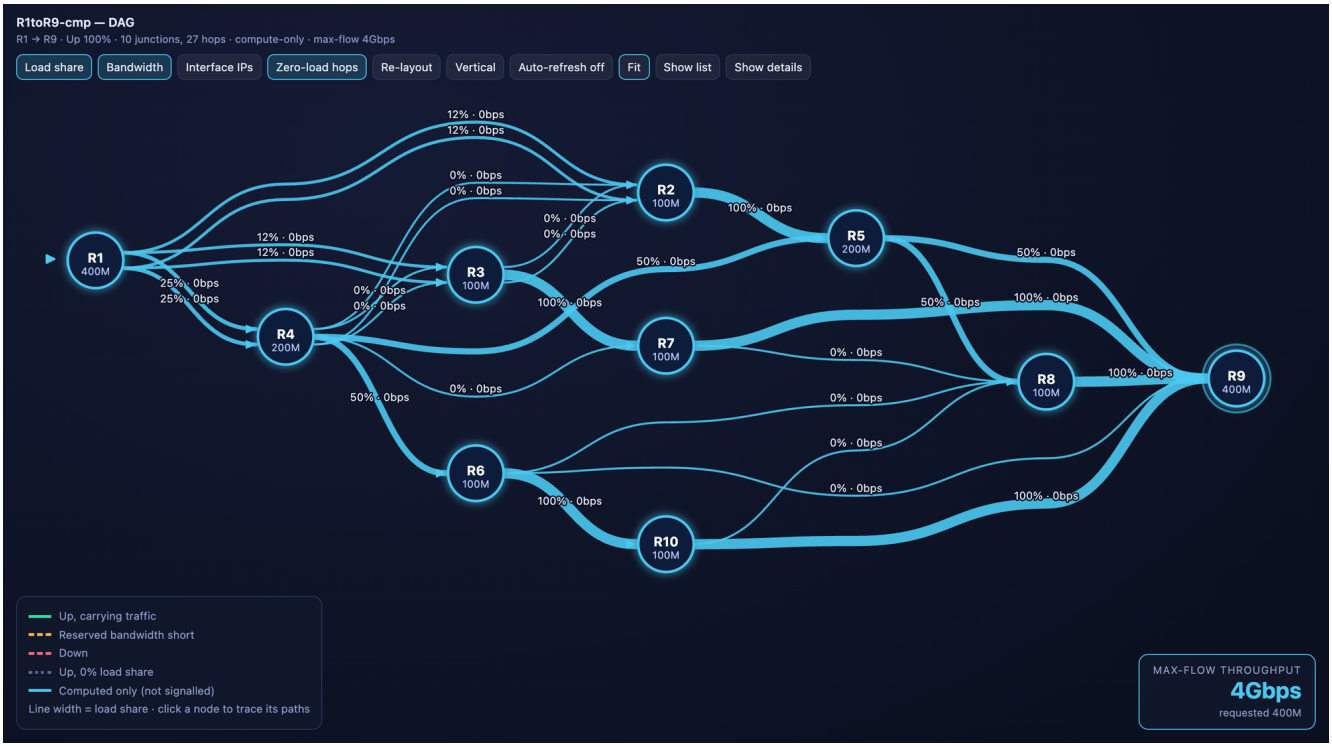


*Figure 5. The same DAG with the Bandwidth view on: each edge shows its load share and its reserved bandwidth. Every edge reserves 0 bps.*

| | Before | With 400m | After removal |
|---|---|---|---|
| Ingress BW | 0bps | 400Mbps | 0bps |
| DAG version | 1 | 2 | 3 |
| Load shares on hops | 0% | 0, 12, 25, 50, 100% | 0% |
| Reserved BW (ResBW) | 0bps | 0bps | 0bps |
| Max-flow | 4Gbps | 4Gbps | 4Gbps |

**Takeaway:** The tunnel now shows how the 400 Mbps would be split across the hops, but nothing is reserved (ResBW is 0bps on every hop) and the junctions are still Down. Bandwidth is only reserved when a tunnel is signaled, which you will do in the next section.

Remove the bandwidth request again and check that the tunnel is back to how it was:

```
configure
delete protocols mpte tunnel R1toR9-cmp ingress 10.123.123.1 bandwidth
show | compare
commit
exit
```

```
[edit protocols mpte tunnel R1toR9-cmp ingress 10.123.123.1]
-     bandwidth 400m;
```

```
show mpte tunnel name R1toR9-cmp extensive
```

```
R1toR9-cmp, Originator: 10.123.123.1, ID: 100
  Up, Availability: 100%, Type: MPLS-Sig
  ComputeEngine: 10.123.123.1, Compute-only
  Ingresses (1):
    10.123.123.1, BW 0bps
  Egresses (1):
    10.123.123.9
  OptimizationObjective: TE, MetricMargin: Infinite
  Metric Min 30 Max 40
  Max-Flow Throughput: 4Gbps
  DAGVersion 3
    Junctions (10):
```

### Ask the LLM

**Config change:** These prompts change the router configuration.

**PROMPT (copy into Copilot chat)**

On router r1, add a bandwidth request of 400m to the ingress of tunnel R1toR9-cmp. Then show me the load share of each next hop at the ingress junction R1, and tell me whether any bandwidth is reserved.

**A good answer includes:** R1 has six next hops with load shares of about 12%, 12%, 12%, 12%, 25% and 25% (two links each to R2, R3 and R4). The reserved bandwidth is 0bps on every hop: the tunnel is compute-only, so nothing is reserved.

**PROMPT (copy into Copilot chat)**

Now remove the bandwidth request from R1toR9-cmp and confirm that the tunnel is back to BW 0bps.

**A good answer includes:** The ingress shows BW 0bps again, all load shares are 0%, and the DAG version has increased.

## 2.3 Compute-only with a metric margin: R1toR9-margin-0-cmp

A metric margin limits the DAG to paths whose metric is close to the shortest. By default the margin is Infinite and every path is allowed. With a margin of 0, only the shortest-metric paths remain. This tunnel combines a margin of 0 with compute-only. Its configuration:

```
set protocols mpte tunnel R1toR9-margin-0-cmp ingress 10.123.123.1
set protocols mpte tunnel R1toR9-margin-0-cmp egress 10.123.123.9
set protocols mpte tunnel R1toR9-margin-0-cmp optimize metric-margin 0
set protocols mpte tunnel R1toR9-margin-0-cmp compute-only
```

The tunnel already exists; you only need to activate it:

```
configure
activate protocols mpte tunnel R1toR9-margin-0-cmp
show | compare
commit
exit
```

```
[edit protocols mpte]
!     active: tunnel R1toR9-margin-0-cmp { ... }
```

**Undo:** run deactivate protocols mpte tunnel R1toR9-margin-0-cmp and commit.

```
show mpte tunnel
```

```
Originator       ID     State  Avail  JCTs Type       SigType TunnelName
10.123.123.1     100    Up     100%   10   MPLS-Sig   Compute R1toR9-cmp
10.123.123.1     101    Up     100%   8    MPLS-Sig   Compute R1toR9-margin-0-cmp

Total: 2
Up 2, Degraded 0, Down 0
```

```
show mpte tunnel name R1toR9-margin-0-cmp extensive
```

```
R1toR9-margin-0-cmp, Originator: 10.123.123.1, ID: 101
  Up, Availability: 100%, Type: MPLS-Sig
  ComputeEngine: 10.123.123.1, Compute-only
  Ingresses (1):
    10.123.123.1, BW 0bps
  Egresses (1):
    10.123.123.9
  OptimizationObjective: TE, MetricMargin: 0
  Metric Min 30 Max 30
  Max-Flow Throughput: 3Gbps
  DAGVersion 1
    Junctions (8):
    ...
    2 Oct 18 18:02:00.480 Configuration Changed
    1 Oct 18 18:02:00.472 Tunnel Created
  Created: Sun Oct 18 18:02:00 2026
Total: 1
Up 1, Degraded 0, Down 0
```

*Figure 6. DAG of R1toR9-margin-0-cmp: only the shortest-metric paths remain. R8 and R10 have dropped out.*

| | R1toR9-cmp (margin Infinite) | R1toR9-margin-0-cmp (margin 0) |
|---|---|---|
| Metric margin | Infinite | 0 |
| Metric from R1 to R9 (min to max) | 30 to 40 | 30 to 30 |
| Junctions | 10 | 8 |
| Hops | 27 | 14 |
| Max-flow throughput | 4Gbps | 3Gbps |

**Takeaway:** R8 and R10 are gone from the DAG, because every path through them has a higher metric than the shortest path. The DAG gets smaller and the max-flow drops from 4 Gbps to 3 Gbps. A smaller margin means fewer, shorter paths; Infinite means all paths.

### Ask the LLM

**Config change:** This prompt changes the router configuration.

**PROMPT (copy into Copilot chat)**

On router r1, activate the MPTE tunnel R1toR9-margin-0-cmp and compare it with R1toR9-cmp: number of junctions, number of hops, max-flow throughput and metric range. Which routers dropped out of the DAG, and why?

**A good answer includes:** R1toR9-cmp has 10 junctions, 27 hops and 4Gbps; R1toR9-margin-0-cmp has 8 junctions, 14 hops and 3Gbps. R8 and R10 dropped out: with a metric margin of 0 only the shortest paths (metric 30) are kept, and R8 and R10 are only on longer paths (metric 40).

## Where you are now

At the end of section 2, two compute-only tunnels are active on R1:

```
show mpte tunnel
```

```
Originator       ID     State  Avail  JCTs Type       SigType TunnelName
10.123.123.1     100    Up     100%   10   MPLS-Sig   Compute R1toR9-cmp
10.123.123.1     101    Up     100%   8    MPLS-Sig   Compute R1toR9-margin-0-cmp

Total: 2
Up 2, Degraded 0, Down 0
```

They do not signal anything, so the network is unchanged. In the next section you will activate tunnels that signal, reserve bandwidth and install forwarding state.

# Section 3. Bring up signaled tunnels

A signaled tunnel does everything a compute-only tunnel does and then sets it up in the network. The ingress signals every junction of the DAG with RSVP. Each junction allocates a label, installs forwarding state and, if the tunnel asks for bandwidth, reserves it on its links. Four things change compared with section 2:

- **SigType** is RSVP instead of Compute.
- **Junctions and hops** are Up instead of Down, and carry labels.
- **The ingress and the transit routers** install forwarding state: routes in inet.3 at the ingress and entries in mpls.0 at the transit routers.
- **Bandwidth** is reserved on the links when the tunnel requests it.

**Plan:** The four tunnels in this section all go from R1 to R9 and add one option at a time: nothing, 400 Mbps of bandwidth, 400 Mbps with a metric margin of 10, and a metric margin of 0. The two compute-only tunnels from section 2 stay active.

## 3.1 Signaled, no bandwidth: R1toR9

The simplest signaled tunnel has an ingress and an egress and nothing else:

```
set protocols mpte tunnel R1toR9 ingress 10.123.123.1
set protocols mpte tunnel R1toR9 egress 10.123.123.9
```

It is already configured; activate it:

```
configure
activate protocols mpte tunnel R1toR9
show | compare
commit
exit
```

```
[edit protocols mpte]
!     active: tunnel R1toR9 { ... }
```

**Undo:** run deactivate protocols mpte tunnel R1toR9 and commit.

```
show mpte tunnel
```

```
Originator       ID     State  Avail  JCTs Type       SigType TunnelName
10.123.123.1     100    Up     100%   10   MPLS-Sig   Compute R1toR9-cmp
10.123.123.1     101    Up     100%   8    MPLS-Sig   Compute R1toR9-margin-0-cmp
10.123.123.1     102    Up     100%   10   MPLS-Sig   RSVP    R1toR9

Total: 3
Up 3, Degraded 0, Down 0
```

The new tunnel shows SigType RSVP. Look at it in detail:

```
show mpte tunnel name R1toR9 extensive
```

```
R1toR9, Originator: 10.123.123.1, ID: 102
  Up, Availability: 100%, Type: MPLS-Sig
  ComputeEngine: 10.123.123.1
  SignalingSource: 10.123.123.1, SignalingType: RSVP
  Ingresses (1):
    10.123.123.1, BW 0bps
  Egresses (1):
    10.123.123.9
  OptimizationObjective: TE, MetricMargin: Infinite
  Metric Min 30 Max 40
  Max-Flow Throughput: 4Gbps
  DAGVersion 1
    ...
    Junctions (10):
      JCT 10.123.123.1, Up, ReqBW 0bps
        NHOPs (6):
          NHOP 10.1.1.2 Index 631 Metric 30
            HopVersion 1, Up, OutLabel: 108
              LoadShare 0%, ReqBW 0bps, ResBW 0bps
          NHOP 10.2.1.2 Index 633 Metric 30
            HopVersion 1, Up, OutLabel: 108
              LoadShare 0%, ReqBW 0bps, ResBW 0bps
```

**What changed:** Compare with section 2: the ComputeEngine line no longer says Compute-only, a new SignalingSource line says SignalingType: RSVP, the junction and the hops are Up instead of Down, and each hop has an OutLabel. The hops still have LoadShare 0% and ReqBW 0bps because the tunnel asks for no bandwidth.

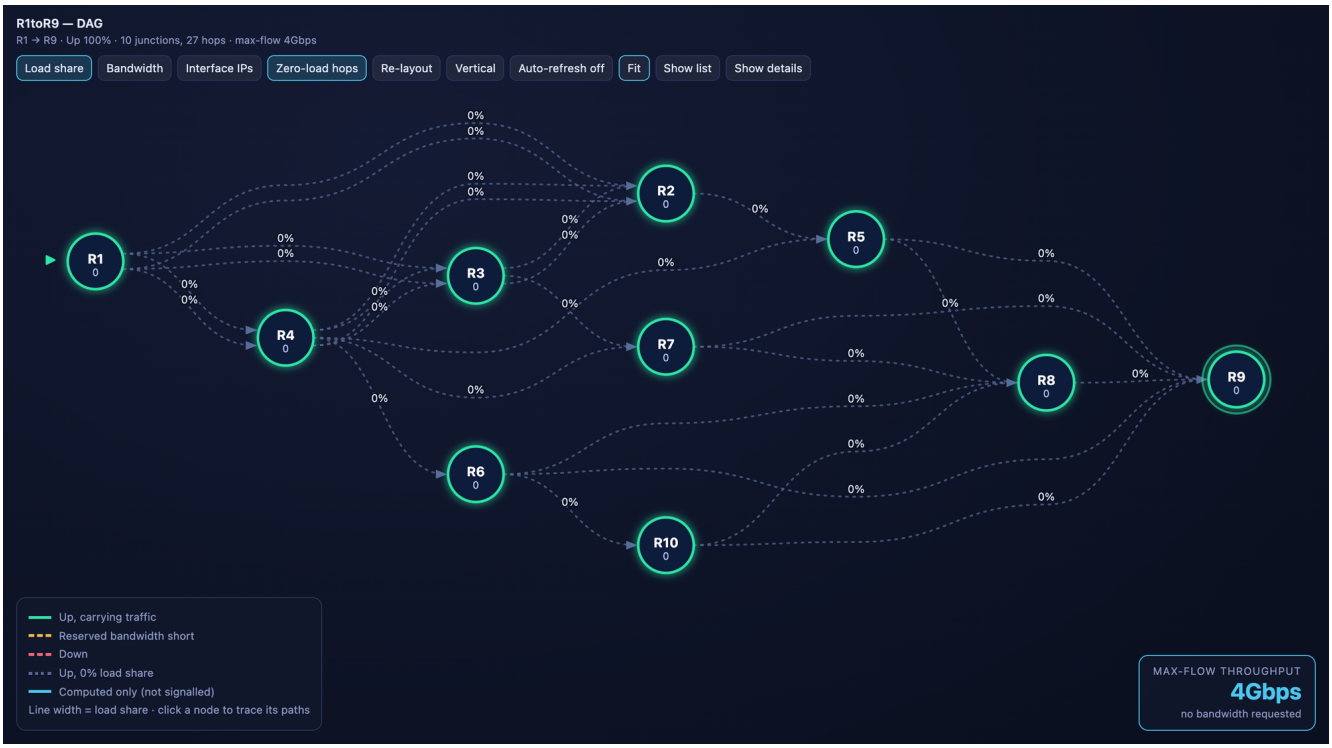


*Figure 7. DAG of R1toR9. Same shape as the compute-only version, but the nodes are green (Up). The dotted edges are hops that are Up with a 0% load share.*

### The tunnel comes up

The tunnel does not appear all at once. Its event log, at the end of the extensive output, shows each junction coming up. While some junctions are still signaling the tunnel is Degraded, and it becomes Up when all of them are up. The whole sequence happens within the same second (the log is newest first):

```
show mpte tunnel name R1toR9 extensive
```

```
    80 Oct 18 18:22:50.134 Up - 100%
    79 Oct 18 18:22:50.134 Junction Up - 10.123.123.10 [10 times since Oct 18 18:22:50]
    78 Oct 18 18:22:50.072 Degraded - 10%
    77 Oct 18 18:22:50.072 Junction Degraded - 10.123.123.10
    76 Oct 18 18:22:50.072 Junction Up - 10.123.123.6 [9 times since Oct 18 18:22:50]
    75 Oct 18 18:22:50.072 Degraded - 20%
    74 Oct 18 18:22:50.072 Junction Up - 10.123.123.7 [3 times since Oct 18 18:22:50]
    73 Oct 18 18:22:50.072 Junction Degraded - 10.123.123.6
    ...
  Created: Sun Oct 18 18:22:48 2026
```

### Forwarding state at the ingress

The ingress now has a route to R9 through the tunnel, with a label to push on every next hop. A compute-only tunnel installs nothing here.

```
show route table inet.3
```

```
inet.3: 1 destinations, 1 routes (1 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both

10.123.123.9/32    *[RSVP/7/1] 00:00:34, metric 1
                       to 10.1.3.4 via eth3, Push 194
                     > to 10.2.3.4 via eth6, Push 194
                       to 10.1.2.3 via eth2, Push 186
                       to 10.2.2.3 via eth5, Push 186
                       to 10.1.1.2 via eth1, Push 108
                       to 10.2.1.2 via eth4, Push 108
```

Labels are assigned by the router at the other end of each link, so all links to the same neighbor use the same label: R2 108, R3 186, R4 194. Your label numbers will differ.

### RSVP sessions and junctions

```
show rsvp mpte session
```

```
Local Ingress MPTE: 1 sessions, 1 up, 0 down, 0 degraded
Local Transit MPTE: 0 sessions, 0 up, 0 down, 0 degraded
Local Egress MPTE: 0 sessions, 0 up, 0 down, 0 degraded
Remote Ingress MPTE: 0 sessions, 0 up, 0 down, 0 degraded
Remote Transit MPTE: 8 sessions, 8 up, 0 down, 0 degraded
Remote Egress MPTE: 1 sessions, 1 up, 0 down, 0 degraded
Total 10 MPTE sessions, 10 up, 0 down, 0 degraded
```

R1 is the signaling source, so it holds the one Local Ingress session and also reports the sessions of the other junctions: 8 transit and 1 egress. That is one session per junction, 10 in all.

```
show mpte junction
```

```
Junction-ID      Originator       ID     Version SigSource        State  Avail  TunnelName
10.123.123.1     10.123.123.1     102    1       10.123.123.1     Up     100%   R1toR9
10.123.123.2     10.123.123.1     102    1       10.123.123.1     Up     100%   R1toR9
10.123.123.3     10.123.123.1     102    1       10.123.123.1     Up     100%   R1toR9
10.123.123.4     10.123.123.1     102    1       10.123.123.1     Up     100%   R1toR9
    ...
Total: 10
Up 10, Degraded 0, Down 0
```

### A transit junction: R5

Run the same commands on a transit router. Log in to R5 the same way as for R1. A transit router knows only its own junction:

```
show mpte junction
```

```
Junction-ID      Originator       ID     Version SigSource        State  Avail  TunnelName
10.123.123.5     10.123.123.1     102    1       10.123.123.1     Up     100%   R1toR9

Total: 1
Up 1, Degraded 0, Down 0
```

R5 forwards the tunnel by label. The label in this run is 126:

```
show route table mpls.0 label 126
```

```
mpls.0: 6 destinations, 6 routes (6 active, 0 holddown, 0 hidden)
126                *[RSVP/7/1] 00:01:55, metric 1
                    >  to 10.1.12.8 via eth3, Swap 89
                       to 10.1.15.9 via eth4, Pop
126(S=0)           *[RSVP/7/1] 00:01:55, metric 1
                    >  to 10.1.12.8 via eth3, Swap 89
                       to 10.1.15.9 via eth4, Pop
```

Read it as: a packet that arrives with label 126 is either swapped to label 89 and sent to 10.1.12.8 (R8) on eth3, or has its label removed (Pop) and sent to 10.1.15.9 (R9) on eth4. Each of those is one of the next hops of R5 in the DAG. The same label also appears with (S=0), for packets that carry more labels underneath.

*Figure 8. Clicking R5 in the page highlights the paths through it and lists its previous hops (from R2 and R4) and next hops (to R8 and R9).*

## Ask the LLM

**Config change:** This prompt changes the router configuration. Undo: ask the LLM to deactivate the tunnel.

**PROMPT (copy into Copilot chat)**

On router r1, activate the MPTE tunnel R1toR9. Is it signaled or compute-only, how many junctions are Up, and how many RSVP sessions does it have?

**A good answer includes:** The tunnel is Up with 100% availability and SigType RSVP, with 10 junctions Up. There are 10 RSVP sessions: one local ingress session on R1, 8 transit and 1 egress.

**PROMPT (copy into Copilot chat)**

On r1, show the MPLS labels pushed toward R9 for tunnel R1toR9 and the next hop each one goes to. Why are the labels not all the same?

**A good answer includes:** label 108 toward R2, label 186 toward R3, label 194 toward R4. Each neighbor allocates the label it wants to receive for the tunnel (distributed label assignment), so the labels differ per neighbor and are the same on parallel links to the same neighbor.

**PROMPT (copy into Copilot chat)**

On router r5, show how packets of the tunnel R1toR9 are forwarded: which label do they arrive with and what does R5 do with them?

**A good answer includes:** They arrive with label 126. R5 either swaps it to 89 and sends it toward R8, or pops it and sends it toward R9.

## 3.2 Signaled with bandwidth: R1toR9-BW400m

Now ask for 400 Mbps. The bandwidth is part of the ingress statement, as in section 2.2, but this tunnel is signaled, so the bandwidth is reserved:

```
set protocols mpte tunnel R1toR9-BW400m ingress 10.123.123.1 bandwidth 400m
set protocols mpte tunnel R1toR9-BW400m egress 10.123.123.9

configure
activate protocols mpte tunnel R1toR9-BW400m
show | compare
commit
exit
```

```
[edit protocols mpte]
!     active: tunnel R1toR9-BW400m { ... }
```

**Undo:** run deactivate protocols mpte tunnel R1toR9-BW400m and commit.

```
show mpte tunnel name R1toR9-BW400m extensive
```

```
R1toR9-BW400m, Originator: 10.123.123.1, ID: 103
  Up, Availability: 100%, Type: MPLS-Sig
  ComputeEngine: 10.123.123.1
  SignalingSource: 10.123.123.1, SignalingType: RSVP
  Ingresses (1):
    10.123.123.1, BW 400Mbps
  Egresses (1):
    10.123.123.9
  OptimizationObjective: TE, MetricMargin: Infinite
  Metric Min 30 Max 40
  Max-Flow Throughput: 4Gbps
  DAGVersion 1
    ...
    Junctions (10):
      JCT 10.123.123.1, Up, ReqBW 400Mbps
        NHOPs (6):
          NHOP 10.1.1.2 Index 631 Metric 30
            HopVersion 1, Up, OutLabel: 109
              LoadShare 12%, ReqBW 50Mbps, ResBW 50Mbps
          NHOP 10.2.1.2 Index 633 Metric 30
            HopVersion 1, Up, OutLabel: 109
              LoadShare 12%, ReqBW 50Mbps, ResBW 50Mbps
          NHOP 10.1.3.4 Index 531 Metric 30
            HopVersion 1, Up, OutLabel: 195
              LoadShare 25%, ReqBW 100Mbps, ResBW 100Mbps
```

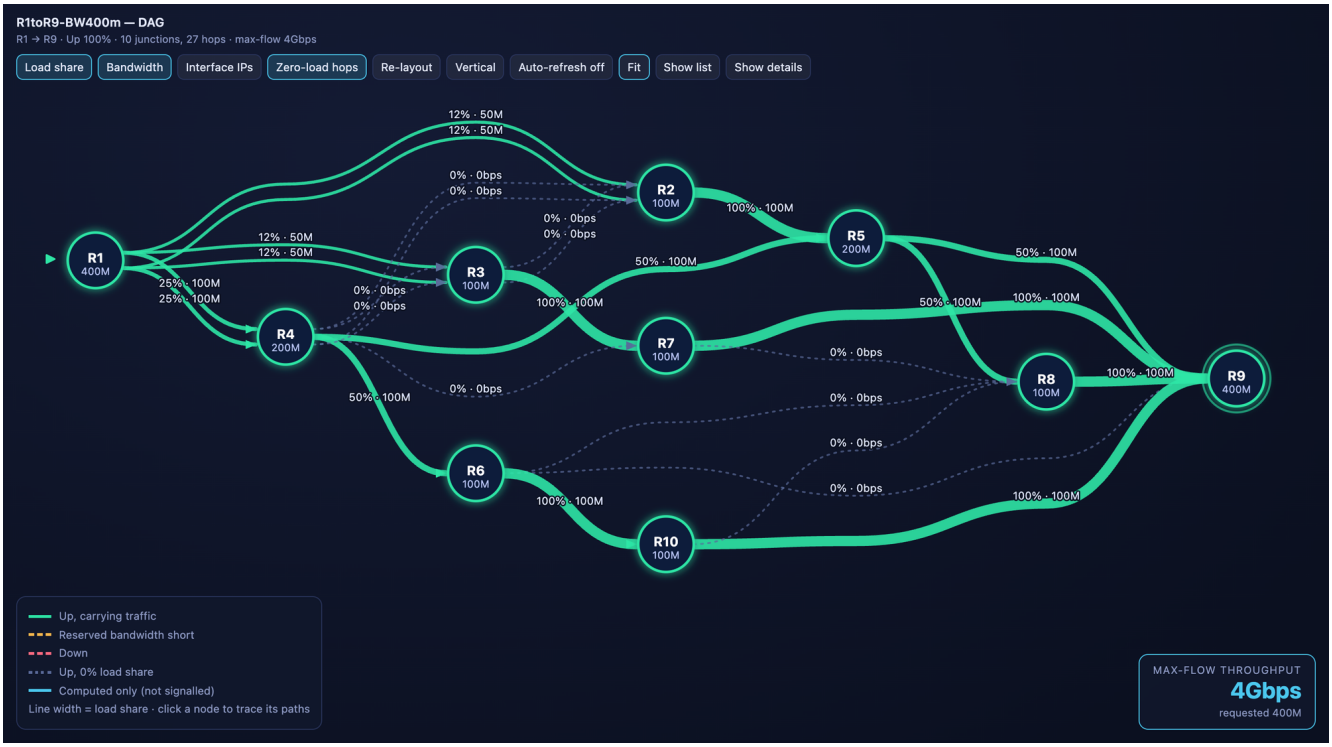


*Figure 9. DAG of R1toR9-BW400m with the Bandwidth view on. Each edge shows its load share and its reserved bandwidth: for example 12% and 50M on the links to R2.*

This is the same split as the compute-only tunnel with 400 Mbps in section 2.2, with one difference: there the reserved bandwidth was 0 bps on every edge, here it equals the requested bandwidth.

## Reserved bandwidth on the links

The reservation shows in the RSVP interface table on R1, which lists the six links to R2, R3 and R4:

```
show rsvp interface
```

```
RSVP interface: 8 active
                       Active  Subscr- Static      Available   Reserved    Highwater
Interface          State  resv    iption  BW          BW          BW          mark
eth0                  Up       0   100%  1000Mbps    1000Mbps    0bps        0bps
eth1                  Up       1   100%  1000Mbps    950Mbps     50Mbps      0bps
eth2                  Up       2   100%  1000Mbps    950Mbps     50Mbps      0bps
eth3                  Up       1   100%  1000Mbps    900Mbps     100Mbps     0bps
eth4                  Up       1   100%  1000Mbps    950Mbps     50Mbps      0bps
eth5                  Up       2   100%  1000Mbps    950Mbps     50Mbps      0bps
eth6                  Up       1   100%  1000Mbps    900Mbps     100Mbps     0bps
lo0.0                 Up       0   100%  0bps        0bps        0bps        0bps
```

| Interface | Neighbor | Static BW | Reserved BW | Available BW |
|---|---|---|---|---|
| eth1 | R2 | 1000Mbps | 50Mbps | 950Mbps |
| eth2 | R3 | 1000Mbps | 50Mbps | 950Mbps |
| eth3 | R4 | 1000Mbps | 100Mbps | 900Mbps |
| eth4 | R2 | 1000Mbps | 50Mbps | 950Mbps |
| eth5 | R3 | 1000Mbps | 50Mbps | 950Mbps |
| eth6 | R4 | 1000Mbps | 100Mbps | 900Mbps |

**Check:** The reservations match the load shares at R1: 12% of 400 Mbps is 50 Mbps on each link to R2 and R3, and 25% is 100 Mbps on each link to R4. Together they add up to the 400 Mbps requested. Before this tunnel, Reserved BW was 0bps on every link.

| | Compute-only with 400m (section 2.2) | Signaled R1toR9-BW400m |
|---|---|---|
| SigType | Compute | RSVP |
| Junctions and hops | Down | Up |
| Reserved BW on the hops (ResBW) | 0bps | 50M and 100M, as requested |
| Reserved BW on R1's links | 0bps | 50M or 100M per link, 400M in total |
| Max-flow | 4Gbps | 4Gbps |

**Takeaway:** Reserving bandwidth takes it away from other tunnels. Every tunnel you bring up after this one sees less available bandwidth on these links.

### Ask the LLM

**Config change:** This prompt changes the router configuration.

**PROMPT (copy into Copilot chat)**

On router r1, activate the MPTE tunnel R1toR9-BW400m. Then show the reserved bandwidth on each of r1's RSVP interfaces and explain how it relates to the load share of each next hop.

**A good answer includes:** Reserved bandwidth of 50 Mbps on eth1 and eth4 (to R2), 50 Mbps on eth2 and eth5 (to R3) and 100 Mbps on eth3 and eth6 (to R4), 400 Mbps in total. This matches the load shares of 12%, 12%, 12%, 12%, 25% and 25% of 400 Mbps.

**PROMPT (copy into Copilot chat)**

What is the difference between requesting 400 Mbps on a compute-only tunnel and on a signaled tunnel?

**A good answer includes:** Both compute the same split of the 400 Mbps across the hops. The compute-only tunnel reserves nothing (ResBW 0bps, junctions Down), while the signaled tunnel reserves the bandwidth on every link it uses.

## 3.3 Signaled with bandwidth and a metric margin: R1toR9-BW400m-margin-10

Now combine bandwidth with a metric margin of 10. The margin limits the DAG to paths that are close to the shortest:

```
set protocols mpte tunnel R1toR9-BW400m-margin-10 ingress 10.123.123.1 bandwidth 400m
set protocols mpte tunnel R1toR9-BW400m-margin-10 egress 10.123.123.9
set protocols mpte tunnel R1toR9-BW400m-margin-10 optimize metric-margin 10

configure
activate protocols mpte tunnel R1toR9-BW400m-margin-10
show | compare
commit
exit
```

```
[edit protocols mpte]
!     active: tunnel R1toR9-BW400m-margin-10 { ... }
```

**Undo:** run deactivate protocols mpte tunnel R1toR9-BW400m-margin-10 and commit.

```
show mpte tunnel name R1toR9-BW400m-margin-10 extensive
```

```
R1toR9-BW400m-margin-10, Originator: 10.123.123.1, ID: 104
  Up, Availability: 100%, Type: MPLS-Sig
  ComputeEngine: 10.123.123.1
  SignalingSource: 10.123.123.1, SignalingType: RSVP
  Ingresses (1):
    10.123.123.1, BW 400Mbps
  Egresses (1):
    10.123.123.9
  OptimizationObjective: TE, MetricMargin: 10
  Metric Min 30 Max 40
  Max-Flow Throughput: 2.8Gbps
  DAGVersion 1
```

*Figure 10. DAG of R1toR9-BW400m-margin-10. R8 has dropped out of the DAG, and the 400 Mbps is split differently (the number in each node is the bandwidth requested at that junction).*

| | R1toR9-BW400m | R1toR9-BW400m-margin-10 |
|---|---|---|
| Metric margin | Infinite | 10 |
| Junctions | 10 | 9 |
| Hops | 27 | 20 |
| Routers missing from the DAG | none | R8 |
| Max-flow | 4Gbps | 2.8Gbps |

**What changed:** The margin removes R8 from the DAG, so the DAG has fewer junctions and hops, and the 400 Mbps is split differently: at R1 it is spread almost evenly over the six links (16% or 17% each) instead of 12% on four links and 25% on two. The max-flow is lower too. This tunnel was also computed after R1toR9-BW400m had reserved bandwidth on the same links, which leaves less bandwidth available.

### Ask the LLM

**Config change:** This prompt changes the router configuration.

**PROMPT (copy into Copilot chat)**

On router r1, activate the MPTE tunnel R1toR9-BW400m-margin-10 and compare it with R1toR9-BW400m: number of junctions, number of hops, max-flow throughput and the router that is missing from the DAG.

**A good answer includes:** R1toR9-BW400m has 10 junctions, 27 hops and 4Gbps; R1toR9-BW400m-margin-10 has 9 junctions, 20 hops and 2.8Gbps. R8 is missing from the second DAG.

## 3.4 Signaled with a metric margin of 0: R1toR9-margin-0

Finally, a signaled tunnel with a margin of 0 and no bandwidth. In section 2.3 you saw the compute-only version of this tunnel:

```
set protocols mpte tunnel R1toR9-margin-0 ingress 10.123.123.1
set protocols mpte tunnel R1toR9-margin-0 egress 10.123.123.9
set protocols mpte tunnel R1toR9-margin-0 optimize metric-margin 0

configure
activate protocols mpte tunnel R1toR9-margin-0
show | compare
commit
exit
```

```
[edit protocols mpte]
!     active: tunnel R1toR9-margin-0 { ... }
```

**Undo:** run deactivate protocols mpte tunnel R1toR9-margin-0 and commit.

```
show mpte tunnel name R1toR9-margin-0 extensive
```

```
R1toR9-margin-0, Originator: 10.123.123.1, ID: 105
  Up, Availability: 100%, Type: MPLS-Sig
  ComputeEngine: 10.123.123.1
  SignalingSource: 10.123.123.1, SignalingType: RSVP
  Ingresses (1):
    10.123.123.1, BW 0bps
  Egresses (1):
    10.123.123.9
  OptimizationObjective: TE, MetricMargin: 0
  Metric Min 30 Max 30
  Max-Flow Throughput: 2.3Gbps
  DAGVersion 1
```

*Figure 11. DAG of R1toR9-margin-0: the same eight routers as the compute-only version, now signaled (green).*

| | R1toR9-margin-0-cmp (section 2.3) | R1toR9-margin-0 (signaled) |
|---|---|---|
| SigType | Compute | RSVP |
| Junctions | 8 | 8 |
| Hops | 14 | 14 |
| Routers missing from the DAG | R8, R10 | R8, R10 |
| Max-flow | 3Gbps | 2.3Gbps |

**Note:** The two DAGs have the same shape. The max-flow of the signaled tunnel is lower. The compute-only tunnel was computed in section 2, before any bandwidth was reserved; this one was computed after the two 400 Mbps tunnels had reserved bandwidth on the links around R1. The most likely explanation is that the max-flow depends on the bandwidth that is available when the DAG is computed.

## 3.5 The network with all four signaled tunnels

Four signaled tunnels and two compute-only tunnels are now active on R1:

```
show mpte tunnel
```

```
Originator       ID     State  Avail  JCTs Type       SigType TunnelName
10.123.123.1     100    Up     100%   10   MPLS-Sig   Compute R1toR9-cmp
10.123.123.1     101    Up     100%   8    MPLS-Sig   Compute R1toR9-margin-0-cmp
10.123.123.1     102    Up     100%   10   MPLS-Sig   RSVP    R1toR9
10.123.123.1     103    Up     100%   10   MPLS-Sig   RSVP    R1toR9-BW400m
10.123.123.1     104    Up     100%   9    MPLS-Sig   RSVP    R1toR9-BW400m-margin-10
10.123.123.1     105    Up     100%   8    MPLS-Sig   RSVP    R1toR9-margin-0

Total: 6
Up 6, Degraded 0, Down 0
```

| Tunnel | SigType | Junctions | Hops | Margin | Requested BW | Max-flow |
|---|---|---|---|---|---|---|
| R1toR9-cmp | Compute | 10 | 27 | Infinite | 0bps | 4Gbps |
| R1toR9-margin-0-cmp | Compute | 8 | 14 | 0 | 0bps | 3Gbps |
| R1toR9 | RSVP | 10 | 27 | Infinite | 0bps | 4Gbps |
| R1toR9-BW400m | RSVP | 10 | 27 | Infinite | 400Mbps | 4Gbps |
| R1toR9-BW400m-margin-10 | RSVP | 9 | 20 | 10 | 400Mbps | 2.8Gbps |
| R1toR9-margin-0 | RSVP | 8 | 14 | 0 | 0bps | 2.3Gbps |

All the signaled tunnels together:

```
show rsvp mpte session
```

```
Local Ingress MPTE: 4 sessions, 4 up, 0 down, 0 degraded
Local Transit MPTE: 0 sessions, 0 up, 0 down, 0 degraded
Local Egress MPTE: 0 sessions, 0 up, 0 down, 0 degraded
Remote Ingress MPTE: 0 sessions, 0 up, 0 down, 0 degraded
Remote Transit MPTE: 29 sessions, 29 up, 0 down, 0 degraded
Remote Egress MPTE: 4 sessions, 4 up, 0 down, 0 degraded
Total 37 MPTE sessions, 37 up, 0 down, 0 degraded
```

```
show route table inet.3
```

```
inet.3: 1 destinations, 4 routes (1 active, 0 holddown, 0 hidden)
+ = Active Route, - = Last Active, * = Both
10.123.123.9/32    *[RSVP/7/1] 09:29:10, metric 1
                       to 10.1.3.4 via eth3, Push 194
                    >  to 10.2.3.4 via eth6, Push 194
                       to 10.1.2.3 via eth2, Push 186
                       to 10.2.2.3 via eth5, Push 186
                       to 10.1.1.2 via eth1, Push 108
                       to 10.2.1.2 via eth4, Push 108
                    [RSVP/7/1] 09:26:28, metric 1
                       to 10.1.3.4 via eth3, Push 195
                    >  to 10.2.3.4 via eth6, Push 195
                       to 10.1.2.3 via eth2, Push 187
                       to 10.2.2.3 via eth5, Push 187
                       to 10.1.1.2 via eth1, Push 109
                       to 10.2.1.2 via eth4, Push 109
                    [RSVP/7/1] 09:25:27, metric 1
                       to 10.1.3.4 via eth3, Push 196
                    >  to 10.2.3.4 via eth6, Push 196
                       to 10.1.2.3 via eth2, Push 188
                       to 10.2.2.3 via eth5, Push 188
                       to 10.1.1.2 via eth1, Push 110
                       to 10.2.1.2 via eth4, Push 110
                    [RSVP/7/1] 00:00:13, metric 1
                       to 10.1.3.4 via eth3, Push 197
                    >  to 10.2.3.4 via eth6, Push 197
                       to 10.1.2.3 via eth2, Push 189
                       to 10.2.2.3 via eth5, Push 189
                       to 10.1.1.2 via eth1, Push 111
                       to 10.2.1.2 via eth4, Push 111
```

There are now four routes to R9, one per signaled tunnel, and one of them is the active route. The compute-only tunnels install none.

### The reserved bandwidth of all tunnels together

Only the two bandwidth tunnels reserve anything (400 Mbps each, 800 Mbps in total), and the reservation is now spread over R1's links:

```
show rsvp interface
```

```
RSVP interface: 8 active
                        Active  Subscr- Static      Available   Reserved    Highwater
Interface          State  resv    iption  BW          BW          BW          mark
eth0                  Up       0   100%  1000Mbps    1000Mbps    0bps        0bps
eth1                  Up       2   100%  1000Mbps    885.714Mbps 114.286Mbps 0bps
eth2                  Up       3   100%  1000Mbps    885.714Mbps 114.286Mbps 0bps
eth3                  Up       2   100%  1000Mbps    828.571Mbps 171.429Mbps 0bps
eth4                  Up       2   100%  1000Mbps    885.714Mbps 114.286Mbps 0bps
eth5                  Up       3   100%  1000Mbps    885.714Mbps 114.286Mbps 0bps
eth6                  Up       2   100%  1000Mbps    828.571Mbps 171.429Mbps 0bps
lo0.0                 Up       0   100%  0bps        0bps        0bps        0bps
```

| Interface | Neighbor | Static BW | Reserved BW | Available BW |
|---|---|---|---|---|
| eth1 | R2 | 1000Mbps | 114.286Mbps | 885.714Mbps |
| eth2 | R3 | 1000Mbps | 114.286Mbps | 885.714Mbps |
| eth3 | R4 | 1000Mbps | 171.429Mbps | 828.571Mbps |
| eth4 | R2 | 1000Mbps | 114.286Mbps | 885.714Mbps |
| eth5 | R3 | 1000Mbps | 114.286Mbps | 885.714Mbps |
| eth6 | R4 | 1000Mbps | 171.429Mbps | 828.571Mbps |

### Ask the LLM

**PROMPT (copy into Copilot chat)**

> Summarize all the MPTE tunnels on router r1 in a table: name, signaling type, number of junctions and requested bandwidth.

> **A good answer includes:** Six tunnels, all Up with 100% availability: R1toR9-cmp and R1toR9-margin-0-cmp are Compute; R1toR9, R1toR9-BW400m, R1toR9-BW400m-margin-10 and R1toR9-margin-0 are RSVP. Junctions: 10, 8, 10, 10, 9 and 8. Requested bandwidth: 400 Mbps for the two BW400m tunnels and none for the others.

**PROMPT (copy into Copilot chat)**

> How much bandwidth is reserved on each of r1's RSVP interfaces, and which tunnels are responsible?

> **A good answer includes:** eth1 114.286Mbps, eth2 114.286Mbps, eth3 171.429Mbps, eth4 114.286Mbps, eth5 114.286Mbps, eth6 171.429Mbps. Only R1toR9-BW400m and R1toR9-BW400m-margin-10 reserve bandwidth, 400 Mbps each.

## Where you are now

At the end of section 3, R1 has two compute-only and four signaled tunnels active. The network now carries RSVP state for four tunnels, with 800 Mbps reserved on the links around R1. In the next sections you will steer tunnels with admin-groups and exclude nodes.

# Section 4. Steer tunnels with admin-groups

In section 1 you saw that every link carries an admin-group, a tag that tunnels can use. So far all the tunnels ignored it and could use every link. A tunnel can instead exclude the links with a tag, or use only the links with a tag. This is how an operator keeps a tunnel off links it should avoid (for example expensive or unreliable links), or on links that are meant for it.

## 4.1 Where the tags come from

The tags are configured per interface on each router, and the TED advertises them. Three groups are defined on every router:

```
show configuration protocols mpls admin-groups
```

```
DONT_GO 1;
MAYBE_GO 2;
JUST_GO 3;
```

Each router tags its own interfaces. On R5 all four links are DONT_GO, and on R6 all four links are MAYBE_GO (the links of R1 and most other routers are JUST_GO). On R5:

```
show configuration protocols mpls | display set | match admin-group
```

```
set protocols mpls interface eth1 admin-group DONT_GO
set protocols mpls interface eth2 admin-group DONT_GO
set protocols mpls interface eth3 admin-group DONT_GO
set protocols mpls interface eth4 admin-group DONT_GO
```

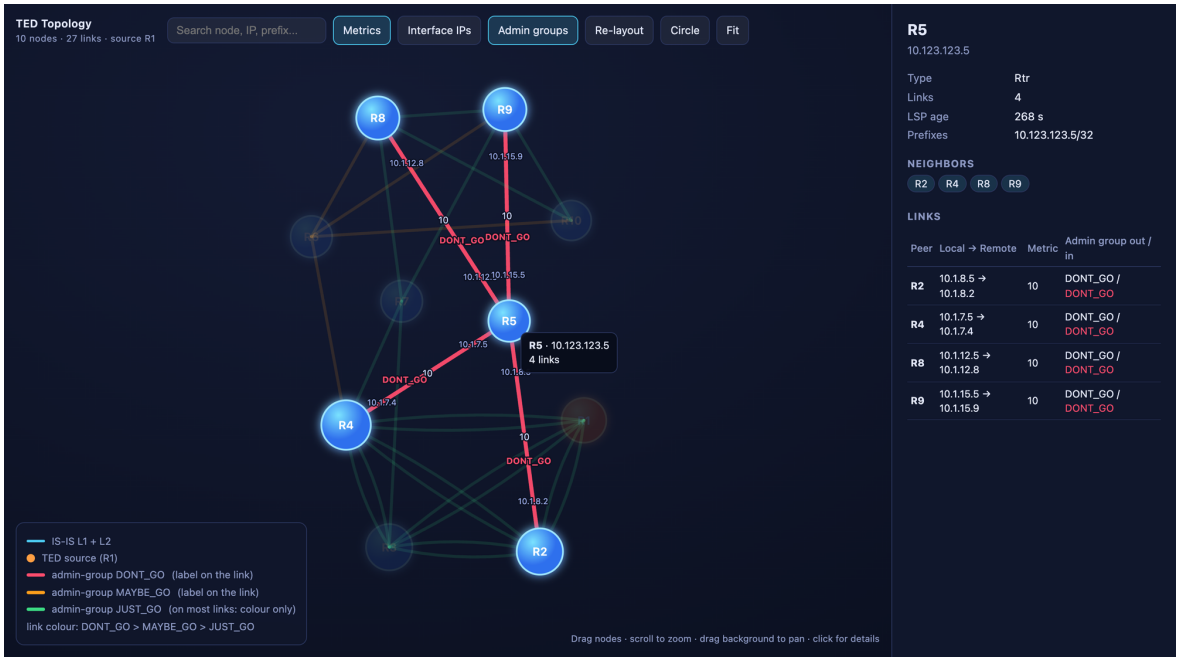


*Figure 12. Click R5 in the topology page: all four of its links (to R2, R4, R8 and R9) are DONT_GO, shown in red. The panel lists the admin-group in each direction.*

| Router | Interfaces | Tag | Links to |
|---|---|---|---|
| R5 | eth1 to eth4 (all of them) | DONT_GO | R2, R4, R8, R9 |
| R6 | eth1 to eth4 (all of them) | MAYBE_GO | R4, R8, R9, R10 |

A link is tagged by the router at each end, and the two ends of each of these eight links carry the same tag (for example R2 tags its link to R5 DONT_GO too). Every other link in the lab is JUST_GO.

## 4.2 Exclude a group: R1toR9-excl-dgo

This tunnel goes from R1 to R9 and must not use any link tagged DONT_GO:

```
set protocols mpte tunnel R1toR9-excl-dgo ingress 10.123.123.1
set protocols mpte tunnel R1toR9-excl-dgo egress 10.123.123.9
set protocols mpte tunnel R1toR9-excl-dgo admin-group exclude DONT_GO

configure
activate protocols mpte tunnel R1toR9-excl-dgo
show | compare
commit
exit
```

```
[edit protocols mpte]
!     active: tunnel R1toR9-excl-dgo { ... }
```

**Undo:** run deactivate protocols mpte tunnel R1toR9-excl-dgo and commit.

```
show mpte tunnel name R1toR9-excl-dgo extensive
```

```
R1toR9-excl-dgo, Originator: 10.123.123.1, ID: 106
  Up, Availability: 100%, Type: MPLS-Sig
  ComputeEngine: 10.123.123.1
  SignalingSource: 10.123.123.1, SignalingType: RSVP
  Ingresses (1):
    10.123.123.1, BW 0bps
  Egresses (1):
    10.123.123.9
  OptimizationObjective: TE, MetricMargin: Infinite
  Metric Min 30 Max 40
  Max-Flow Throughput: 2.42857Gbps
  DAGVersion 1
    Affinities:
      ExcludeAny: DONT_GO
    Junctions (8):
      JCT 10.123.123.1, Up, ReqBW 0bps
```

A new Affinities section shows the rule the DAG was computed with: ExcludeAny: DONT_GO. The tunnel is Up with 8 junctions and 17 hops.

*Figure 13. DAG of R1toR9-excl-dgo. R2 and R5 are missing, and so is every link tagged DONT_GO.*

| | R1toR9 (no tags) | R1toR9-excl-dgo |
|---|---|---|
| Junctions | 10 | 8 |
| Hops | 27 | 17 |
| Routers missing from the DAG | none | R2, R5 |
| Metric from R1 to R9 (min to max) | 30 to 40 | 30 to 40 |
| Max-flow | 4Gbps | 2.42857Gbps |

**Why R2 and R5:** R5 is missing because all of its links are DONT_GO. R2 is missing as a side effect: in the full DAG its next hops were R5, and with R5 gone it would lead only to R3 and R4, which R1 already reaches directly, so it adds no path. The max-flow drops because the capacity that went through R5 is gone.

## Check it on the routers

A router that is not part of the DAG is not a junction of the tunnel. R5 is a junction of the four earlier tunnels but not of the new one; R6 is a junction of the new one:

```
show mpte junction
```

| Router | Junction of these tunnels |
|---|---|
| R2 | R1toR9, R1toR9-BW400m, R1toR9-BW400m-margin-10, R1toR9-margin-0 |
| R5 | R1toR9, R1toR9-BW400m, R1toR9-BW400m-margin-10, R1toR9-margin-0 |
| R6 | R1toR9, R1toR9-BW400m, R1toR9-BW400m-margin-10, R1toR9-margin-0, R1toR9-excl-dgo |
| R8 | R1toR9, R1toR9-BW400m, R1toR9-excl-dgo, R1toR9-incl-jgo |

Run the command on R5 and R6 yourself and compare the lists.

## Ask the LLM

**Config change:** This prompt changes the router configuration. Undo: ask the LLM to deactivate the tunnel.

**PROMPT (copy into Copilot chat)**

> On router r1, activate the MPTE tunnel R1toR9-excl-dgo. Which routers are in its DAG compared with R1toR9, and why are R2 and R5 missing?

**A good answer includes:** The DAG has 8 junctions (R1, R3, R4, R6, R7, R8, R9, R10); R2 and R5 are missing. The tunnel excludes the admin-group DONT_GO, and all four links of R5 are tagged DONT_GO, so R5 cannot be used. R2 is missing too: with R5 gone it would lead only to R3 and R4, which R1 already reaches directly.

**PROMPT (copy into Copilot chat)**

> Which links in the topology are tagged DONT_GO, and which tunnels on r1 avoid them?

**A good answer includes:** The links R2-R5, R4-R5, R5-R8 and R5-R9. The tunnel R1toR9-excl-dgo excludes them by configuration (admin-group exclude DONT_GO); the other tunnels do not use admin-groups and may use them.

## 4.3 Include a group: R1toR9-incl-jgo

The opposite rule: this tunnel may only use links tagged JUST_GO.

```
set protocols mpte tunnel R1toR9-incl-jgo ingress 10.123.123.1
set protocols mpte tunnel R1toR9-incl-jgo egress 10.123.123.9
set protocols mpte tunnel R1toR9-incl-jgo admin-group include-any JUST_GO

configure
activate protocols mpte tunnel R1toR9-incl-jgo
show | compare
commit
exit
```

```
[edit protocols mpte]
!     active: tunnel R1toR9-incl-jgo { ... }
```

**Undo:** run deactivate protocols mpte tunnel R1toR9-incl-jgo and commit.

```
show mpte tunnel name R1toR9-incl-jgo extensive
```

```
R1toR9-incl-jgo, Originator: 10.123.123.1, ID: 107
  Up, Availability: 100%, Type: MPLS-Sig
  ComputeEngine: 10.123.123.1
  SignalingSource: 10.123.123.1, SignalingType: RSVP
  Ingresses (1):
    10.123.123.1, BW 0bps
  Egresses (1):
    10.123.123.9
  OptimizationObjective: TE, MetricMargin: Infinite
  Metric Min 30 Max 50
  Max-Flow Throughput: 1.77143Gbps
  DAGVersion 1
    Affinities:
      IncludeAny: JUST_GO
    Junctions (7):
      JCT 10.123.123.1, Up, ReqBW 0bps
```

The rule now reads IncludeAny: JUST_GO: a link qualifies if it carries at least one of the listed groups. Here only JUST_GO is listed.

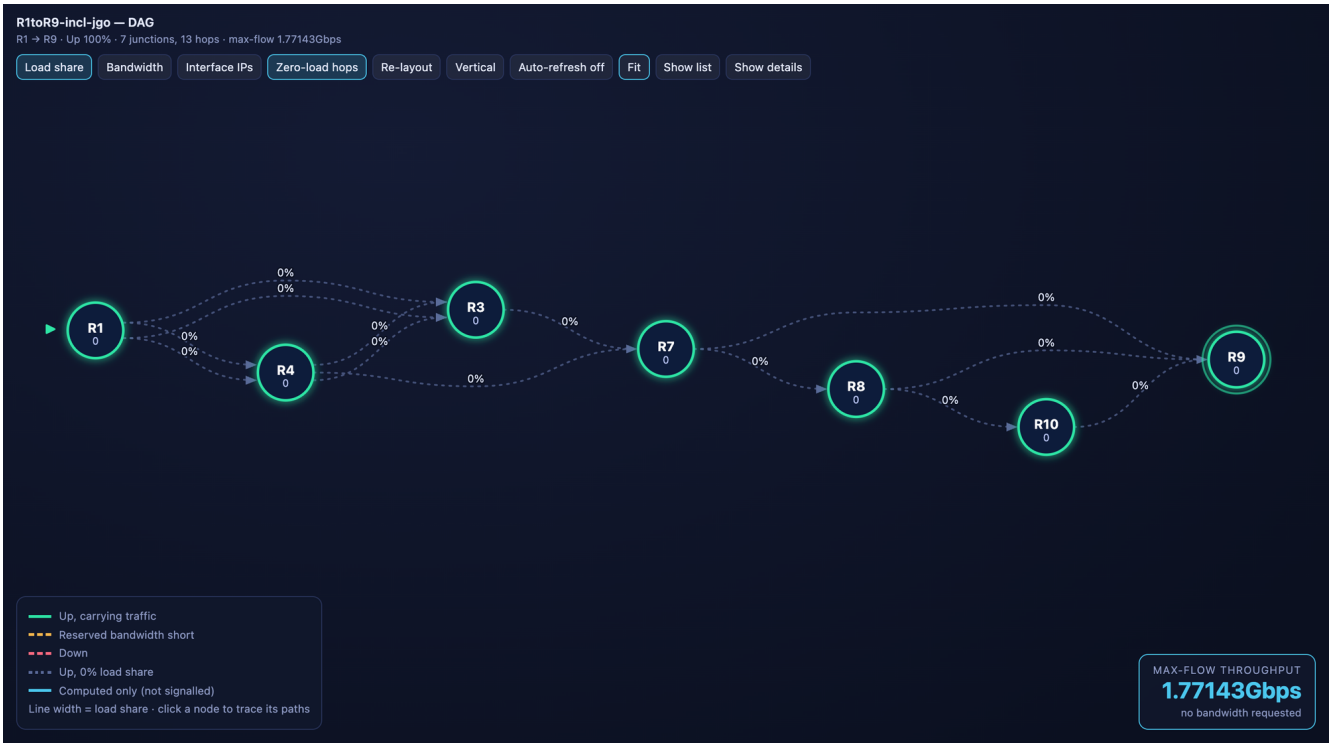


*Figure 14. DAG of R1toR9-incl-jgo. Only JUST_GO links remain, so R2, R5 and R6 are missing.*

| | R1toR9 | R1toR9-excl-dgo | R1toR9-incl-jgo |
|---|---|---|---|
| Rule | none | exclude DONT_GO | include-any JUST_GO |
| Junctions | 10 | 8 | 7 |
| Hops | 27 | 17 | 13 |
| Routers missing | none | R2, R5 | R2, R5, R6 |
| Metric from R1 to R9 | 30 to 40 | 30 to 40 | 30 to 50 |
| Max-flow | 4Gbps | 2.42857Gbps | 1.77143Gbps |

**Compare:** Excluding DONT_GO removes R5 and R2. Including only JUST_GO removes R5 and R2 as well and R6, whose links are all MAYBE_GO, so the DAG is smaller still. Because the paths are fewer and longer, the longest metric rises to 50 and the max-flow drops to about 1.8 Gbps. Use exclude to keep a tunnel off specific links and include to keep it on a specific set of links.

## Ask the LLM

**Config change:** This prompt changes the router configuration.

**PROMPT (copy into Copilot chat)**

On router r1, activate the MPTE tunnel R1toR9-incl-jgo and compare its DAG with R1toR9-excl-dgo: which routers does each one use, and which one is more restrictive?

**A good answer includes:** R1toR9-excl-dgo has 8 junctions and R1toR9-incl-jgo has 7. Both drop R2 and R5; R1toR9-incl-jgo also drops R6, because all links of R6 are MAYBE_GO and the tunnel accepts only JUST_GO. The include tunnel is more restrictive, with max-flow of about 1.8 Gbps against 2.4 Gbps.

**PROMPT (copy into Copilot chat)**

If I wanted a tunnel from R1 to R9 that may use JUST_GO and MAYBE_GO links but never DONT_GO links, which admin-group option would I use?

> **A good answer includes:** Either admin-group exclude DONT_GO (as in R1toR9-excl-dgo), or admin-group include-any with both JUST_GO and MAYBE_GO. The exclude form is simpler.

## Where you are now

At the end of section 4, R1 has eight tunnels active: two compute-only and six signaled.

```
show mpte tunnel
```

```
Originator       ID     State  Avail  JCTs Type       SigType TunnelName
10.123.123.1     100    Up     100%   10   MPLS-Sig   Compute R1toR9-cmp
10.123.123.1     101    Up     100%   8    MPLS-Sig   Compute R1toR9-margin-0-cmp
10.123.123.1     102    Up     100%   10   MPLS-Sig   RSVP    R1toR9
10.123.123.1     103    Up     100%   10   MPLS-Sig   RSVP    R1toR9-BW400m
10.123.123.1     104    Up     100%   9    MPLS-Sig   RSVP    R1toR9-BW400m-margin-10
10.123.123.1     105    Up     100%   8    MPLS-Sig   RSVP    R1toR9-margin-0
10.123.123.1     106    Up     100%   8    MPLS-Sig   RSVP    R1toR9-excl-dgo
10.123.123.1     107    Up     100%   7    MPLS-Sig   RSVP    R1toR9-incl-jgo

Total: 8
Up 8, Degraded 0, Down 0
```

R1 holds 6 ingress RSVP sessions, one per signaled tunnel. None of the admin-group tunnels requested bandwidth, so the reserved bandwidth is unchanged from section 3. In the next section you will keep a tunnel away from specific routers instead of specific links.

# Section 5. Keep a tunnel away from specific routers

Admin-groups, in section 4, act on links: a tag on a link decides whether a tunnel may use it. Sometimes you want to avoid a router instead, for example one that is being maintained or that is overloaded. For that, MPTE lets you define a list of routers (an exclude-hops list) and attach it to a tunnel.

## 5.1 Define the list of routers to avoid

The lab has one list, EXCL_R7_R10, that contains R7 and R10. A hop is identified by the router ID of the router, not by an interface address. The list is defined once, at the top level of the MPTE configuration, and can then be used by any tunnel:

```
set protocols mpte exclude-hops EXCL_R7_R10 hop 10.123.123.7
set protocols mpte exclude-hops EXCL_R7_R10 hop 10.123.123.10
```

The tunnel that uses it asks for 500 Mbps of bandwidth as well, to show that the options combine. It goes from R1 to R8 and refers to the list with the exclude statement:

```
set protocols mpte tunnel R1toR8-BW500m-excl-r7-r10 ingress 10.123.123.1 bandwidth 500m
set protocols mpte tunnel R1toR8-BW500m-excl-r7-r10 egress 10.123.123.8
set protocols mpte tunnel R1toR8-BW500m-excl-r7-r10 exclude EXCL_R7_R10
```

The configuration is already on R1. To see it, run the Junos command inside the device shell and filter it:

```
start shell command "cli -c 'show configuration protocols mpte | display set' | grep -E
'EXCL_R7_R10|R1toR8-BW500m-excl-r7-r10'"
```

```
set protocols mpte exclude-hops EXCL_R7_R10 hop 10.123.123.7
set protocols mpte exclude-hops EXCL_R7_R10 hop 10.123.123.10
set protocols mpte tunnel R1toR8-BW500m-excl-r7-r10 ingress 10.123.123.1 bandwidth 500m
set protocols mpte tunnel R1toR8-BW500m-excl-r7-r10 egress 10.123.123.8
set protocols mpte tunnel R1toR8-BW500m-excl-r7-r10 exclude EXCL_R7_R10
```

**Do not confuse:** exclude (hops) is a different statement from admin-group exclude in section 4: one refers to a list of routers, the other to a tag on links.

## 5.2 First, the same tunnel without the exclusion: R1toR8

To see what the list changes, bring up the plain tunnel to R8 first. It has no bandwidth and no exclusion:

```
configure
activate protocols mpte tunnel R1toR8
show | compare
commit
exit
```

```
[edit protocols mpte]
!     active: tunnel R1toR8 { ... }
```

**Undo:** run deactivate protocols mpte tunnel R1toR8 and commit.

```
show mpte tunnel name R1toR8 extensive
```

```
R1toR8, Originator: 10.123.123.1, ID: 108
  Up, Availability: 100%, Type: MPLS-Sig
  ComputeEngine: 10.123.123.1
  SignalingSource: 10.123.123.1, SignalingType: RSVP
  Ingresses (1):
    10.123.123.1, BW 0bps
  Egresses (1):
    10.123.123.8
  OptimizationObjective: TE, MetricMargin: Infinite
  Metric Min 30 Max 40
  Max-Flow Throughput: 3.2Gbps
  DAGVersion 1
```

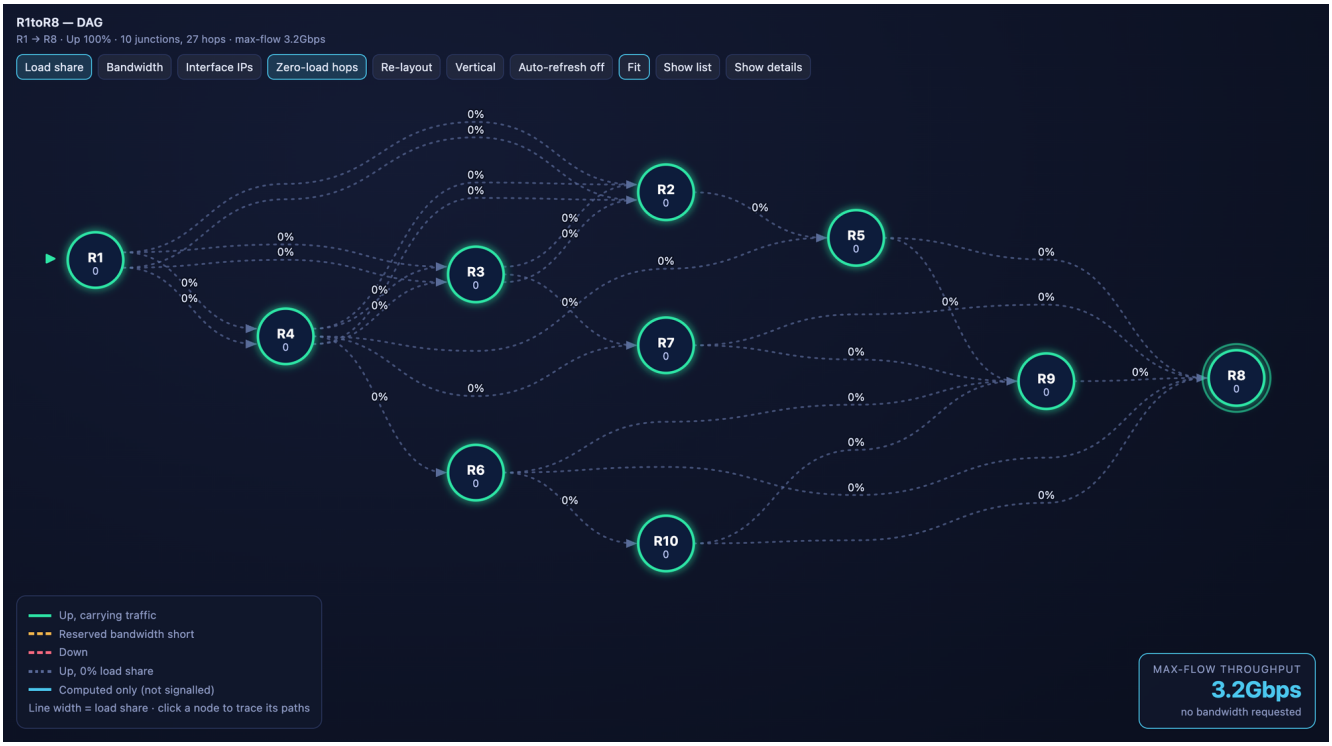


*Figure 15. DAG of R1toR8: 10 junctions and 27 hops. Every router can be used, including R7 and R10.*

## 5.3 Exclude R7 and R10: R1toR8-BW500m-excl-r7-r10

```
configure
activate protocols mpte tunnel R1toR8-BW500m-excl-r7-r10
show | compare
commit
exit
```

```
[edit protocols mpte]
!     active: tunnel R1toR8-BW500m-excl-r7-r10 { ... }
```

**Undo:** run deactivate protocols mpte tunnel R1toR8-BW500m-excl-r7-r10 and commit.

```
show mpte tunnel name R1toR8-BW500m-excl-r7-r10 extensive
```

```
R1toR8-BW500m-excl-r7-r10, Originator: 10.123.123.1, ID: 109
  Up, Availability: 100%, Type: MPLS-Sig
  ComputeEngine: 10.123.123.1
  SignalingSource: 10.123.123.1, SignalingType: RSVP
  Ingresses (1):
    10.123.123.1, BW 500Mbps
  Egresses (1):
    10.123.123.8
  OptimizationObjective: TE, MetricMargin: Infinite
  Metric Min 30 Max 40
  Max-Flow Throughput: 2.42857Gbps
  DAGVersion 1
    Exclude: EXCL_R7_R10
    Junctions (8):
```

A new line, Exclude: EXCL_R7_R10, shows which list the DAG was computed with. The tunnel is Up with 8 junctions and 20 hops.

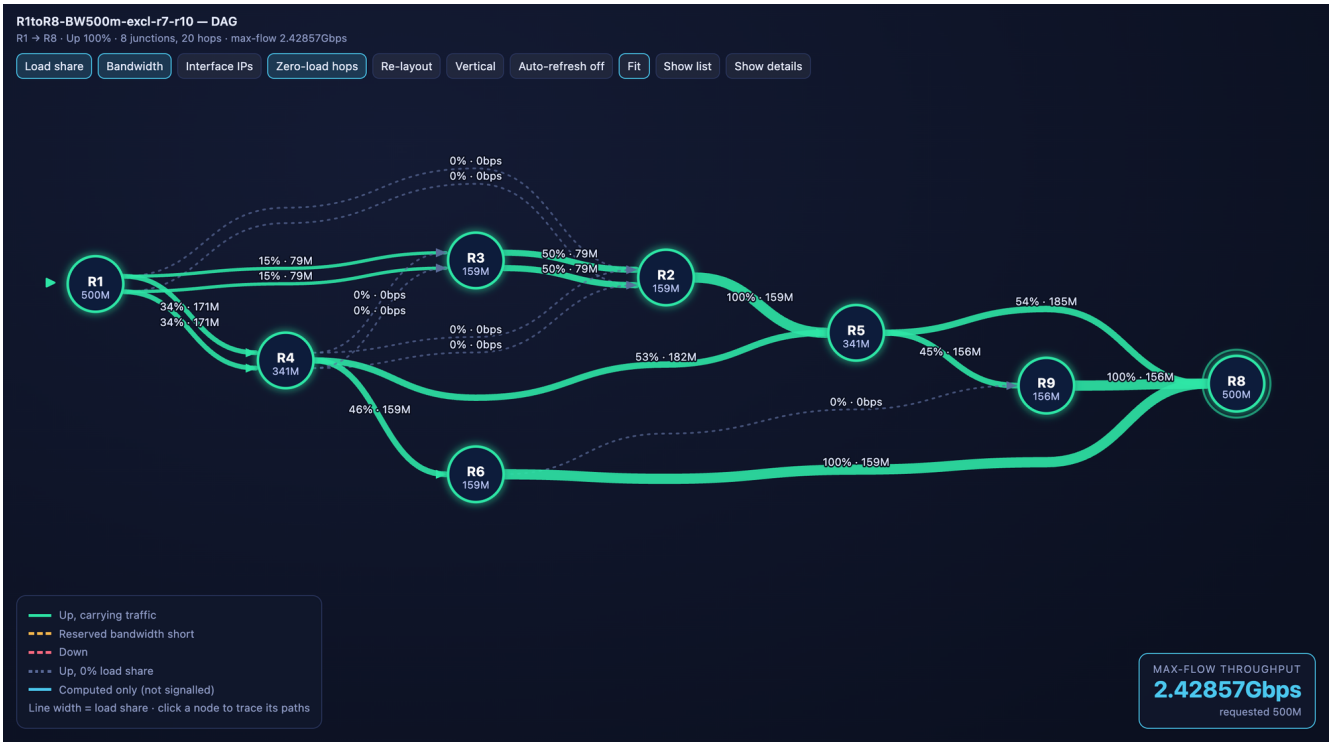


*Figure 16. DAG of R1toR8-BW500m-excl-r7-r10 with the Bandwidth view on. R7 and R10 are gone. The number in each node is the bandwidth requested at that junction; each edge shows its load share and its reserved bandwidth.*

| | R1toR8 | R1toR8-BW500m-excl-r7-r10 |
|---|---|---|
| Exclusion | none | R7 and R10 (EXCL_R7_R10) |
| Requested bandwidth | 0bps | 500Mbps |
| Junctions | 10 | 8 |
| Hops | 27 | 20 |
| Routers missing from the DAG | none | R7, R10 |
| Max-flow | 3.2Gbps | 2.42857Gbps |

**What changed:** The exclusion takes R7 and R10 out of the DAG, and every link of those routers with them. The two tunnels differ in the bandwidth request too, so the lower max-flow is not only the effect of the exclusion. R9 is still in the DAG even though the tunnel ends at R8: R9 is a neighbor of R8 and lies on the paths R5 to R9 to R8, and nothing excludes it. Excluding a router is stronger than excluding links by tag: the router is not used for any path.

## Check it on the routers

The excluded routers are not junctions of the new tunnel. R7 and R10 are junctions of R1toR8, but not of R1toR8-BW500m-excl-r7-r10, while R8 (the egress) and R6 are junctions of both:

```
show mpte junction
```

| Router | Junction of these tunnels |
|---|---|
| R7 | R1toR9, R1toR9-BW400m, R1toR9-BW400m-margin-10, R1toR9-margin-0, R1toR9-excl-dgo, R1toR9-incl-jgo, R1toR8 |
| R10 | R1toR9, R1toR9-BW400m, R1toR9-BW400m-margin-10, R1toR9-excl-dgo, R1toR9-incl-jgo, R1toR8 |
| R8 | R1toR9, R1toR9-BW400m, R1toR9-excl-dgo, R1toR9-incl-jgo, R1toR8, R1toR8-BW500m-excl-r7-r10 |
| R6 | R1toR9, R1toR9-BW400m, R1toR9-BW400m-margin-10, R1toR9-margin-0, R1toR9-excl-dgo, R1toR8, R1toR8-BW500m-excl-r7-r10 |

Run the command on R7 and R10 yourself and look for the new tunnel in the lists.

## The bandwidth that the tunnel reserves

The 500 Mbps request is reserved on top of the 800 Mbps from section 3:

```
show rsvp interface
```

```
RSVP interface: 8 active
                          Active  Subscr- Static      Available   Reserved    Highwater
Interface          State  resv    iption  BW          BW          BW          mark
eth0                   Up       0   100%  1000Mbps    1000Mbps    0bps        0bps
eth1                   Up       2   100%  1000Mbps    885.714Mbps 114.286Mbps 0bps
eth2                   Up       4   100%  1000Mbps    806.302Mbps 193.697Mbps 0bps
eth3                   Up       3   100%  1000Mbps    657.983Mbps 342.017Mbps 0bps
eth4                   Up       2   100%  1000Mbps    885.714Mbps 114.286Mbps 0bps
eth5                   Up       4   100%  1000Mbps    806.302Mbps 193.697Mbps 0bps
eth6                   Up       3   100%  1000Mbps    657.983Mbps 342.017Mbps 0bps
lo0.0                  Up       0   100%  0bps        0bps        0bps        0bps
```

| Interface | Neighbor | Static BW | Reserved BW | Available BW |
|---|---|---|---|---|
| eth1 | R2 | 1000Mbps | 114.286Mbps | 885.714Mbps |
| eth2 | R3 | 1000Mbps | 193.697Mbps | 806.302Mbps |
| eth3 | R4 | 1000Mbps | 342.017Mbps | 657.983Mbps |
| eth4 | R2 | 1000Mbps | 114.286Mbps | 885.714Mbps |
| eth5 | R3 | 1000Mbps | 193.697Mbps | 806.302Mbps |
| eth6 | R4 | 1000Mbps | 342.017Mbps | 657.983Mbps |

**Check:** The six links of R1 now carry about 1300 Mbps of reservations in total: 400 Mbps for R1toR9-BW400m, 400 Mbps for R1toR9-BW400m-margin-10 and 500 Mbps for the new tunnel.

## 5.4 Exclude links or exclude routers?

| | Admin-group (section 4) | Exclude-hops list (section 5) |
|---|---|---|
| What it acts on | Links, by their tag | Routers, by their router ID |
| Configured with | admin-group exclude <group> or include-any <group> | exclude <list>, after defining the list with exclude-hops <list> hop <router-id> |
| Where the tag or list lives | Tag on each interface (mpls) | List under protocols mpte |
| Shown in the extensive output as | Affinities: ExcludeAny / IncludeAny | Exclude: <list> |
| Lab tunnel | R1toR9-excl-dgo, R1toR9-incl-jgo | R1toR8-BW500m-excl-r7-r10 |
| Good for | Keeping tunnels off, or on, a class of links | Avoiding a router, for example during maintenance |

## 5.5 Try it yourself: build your own exclude list

Now create a tunnel from scratch. The tunnel goes from R1 to R9 and must avoid R6. You define the list, then the tunnel that uses it:

```
configure
set protocols mpte exclude-hops EXCL_R6 hop 10.123.123.6
set protocols mpte tunnel R1toR9-excl-r6 ingress 10.123.123.1
set protocols mpte tunnel R1toR9-excl-r6 egress 10.123.123.9
set protocols mpte tunnel R1toR9-excl-r6 exclude EXCL_R6
show | compare
commit
exit
```

```
[edit protocols mpte]
+    exclude-hops EXCL_R6 {
+        hop 10.123.123.6;
+    }
[edit protocols mpte]
+    tunnel R1toR9-excl-r6 {
+        ingress 10.123.123.1;
+        egress 10.123.123.9;
+        exclude EXCL_R6;
+    }
```

```
show mpte tunnel name R1toR9-excl-r6 extensive
```

```
R1toR9-excl-r6, Originator: 10.123.123.1, ID: 110
  Up, Availability: 100%, Type: MPLS-Sig
  ComputeEngine: 10.123.123.1
  SignalingSource: 10.123.123.1, SignalingType: RSVP
  Ingresses (1):
    10.123.123.1, BW 0bps
  Egresses (1):
    10.123.123.9
  OptimizationObjective: TE, MetricMargin: Infinite
  Metric Min 30 Max 50
  Max-Flow Throughput: 2.7Gbps
  DAGVersion 1
    Exclude: EXCL_R6
    Junctions (9):
```

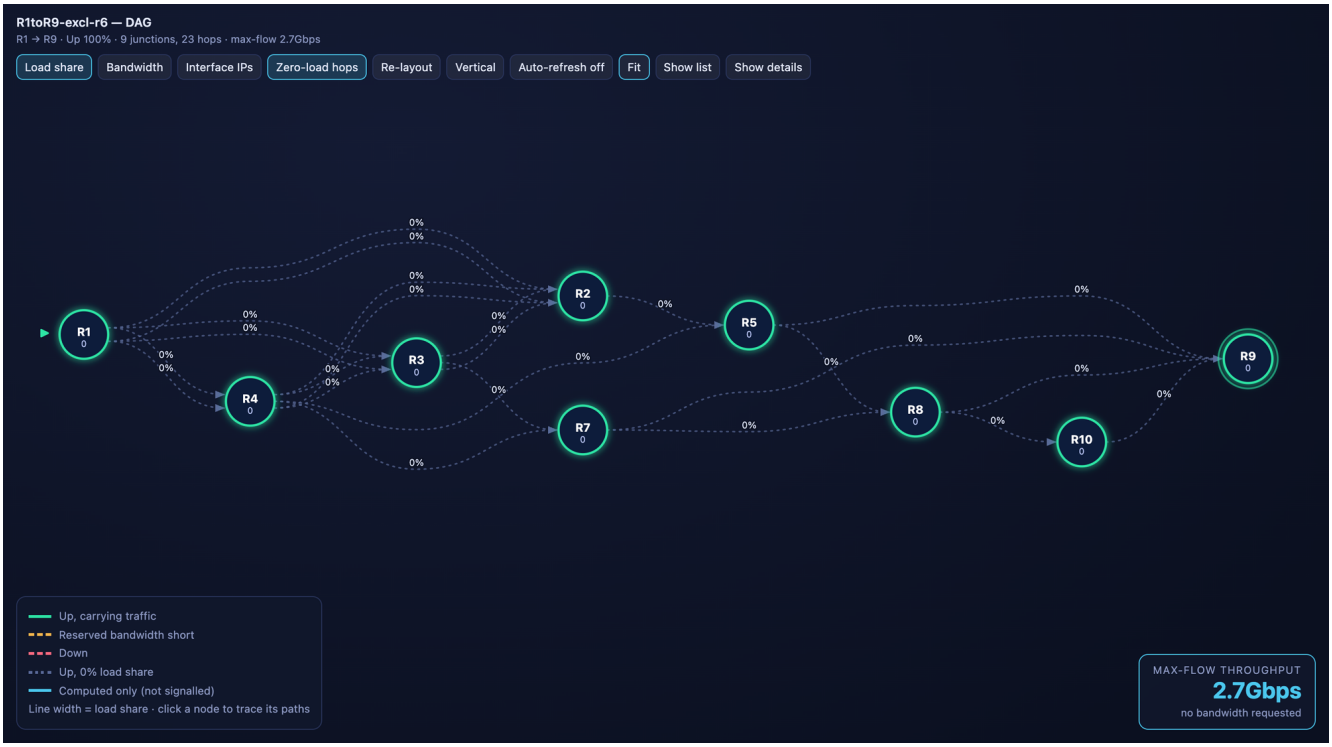


*Figure 17. DAG of your tunnel R1toR9-excl-r6: 9 junctions and 23 hops. R6 is missing.*

| | R1toR9 | R1toR9-excl-r6 (yours) |
|---|---|---|
| Junctions | 10 | 9 |
| Hops | 27 | 23 |
| Routers missing from the DAG | none | R6 |
| Metric from R1 to R9 (min to max) | 30 to 40 | 30 to 50 |
| Max-flow | 4Gbps | 2.7Gbps |

**What you see:** The tunnel avoids R6 and uses the paths through R2, R3, R4, R5, R7, R8 and R10. Because R6 is not available, the longest path has a higher metric than in R1toR9.

Check that R6 is not a junction of your tunnel, but R5 is, by running show mpte junction on both routers.

Clean up when you are done, so the next exercise starts from the same state. Delete the tunnel and the list:

```
configure
delete protocols mpte tunnel R1toR9-excl-r6
delete protocols mpte exclude-hops EXCL_R6
show | compare
commit
exit
```

```
[edit protocols mpte]
-    exclude-hops EXCL_R6 {
-        hop 10.123.123.6;
-    }
-    tunnel R1toR9-excl-r6 {
-        ingress 10.123.123.1;
-        egress 10.123.123.9;
-        exclude EXCL_R6;
-    }
```

**Config change:** this is a real configuration change that adds and then removes statements. Unlike the lab tunnels, which you only activate and deactivate, your tunnel is created and deleted.

## 5.6 Ask the LLM

**Config change:** These prompts change the router configuration.

**PROMPT (copy into Copilot chat)**

On router r1, activate the MPTE tunnel R1toR8-BW500m-excl-r7-r10. Which routers does its DAG contain, which routers are excluded, and which list is used to exclude them?

**A good answer includes:** The DAG has 8 junctions (R1, R2, R3, R4, R5, R6, R8, R9). R7 and R10 are excluded through the exclude-hops list EXCL_R7_R10, which contains their router IDs 10.123.123.7 and 10.123.123.10.

**PROMPT (copy into Copilot chat)**

Compare the tunnels R1toR8 and R1toR8-BW500m-excl-r7-r10 on r1: junctions, hops, requested bandwidth and max-flow throughput. What does each option contribute?

**A good answer includes:** R1toR8 has 10 junctions, 27 hops, no bandwidth request and 3.2Gbps. R1toR8-BW500m-excl-r7-r10 has 8 junctions, 20 hops, a 500 Mbps request and 2.42857Gbps. The exclusion removes R7 and R10 and their links from the DAG; the bandwidth request reserves 500 Mbps on the links it uses.

**PROMPT (copy into Copilot chat)**

Is R9 part of the DAG of R1toR8-BW500m-excl-r7-r10 even though the tunnel ends at R8? Explain.

**A good answer includes:** Yes. R9 is adjacent to R8, so there are paths such as R5 to R9 to R8 and R6 to R9 to R8, and R9 is not on the exclude list.

**PROMPT (copy into Copilot chat)**

On router r1, create an exclude list EXCL_R6 that contains router R6, and a tunnel R1toR9-excl-r6 from R1 to R9 that uses it. Show me its DAG and tell me which routers are missing.

**A good answer includes:** The tunnel is Up with 9 junctions and 23 hops; R6 is the only router missing. (Undo: ask the LLM to delete the tunnel and the list.)

## Where you are now

At the end of section 5, R1 has ten tunnels active: two compute-only and eight signaled.

```
show mpte tunnel
```

```
Originator       ID     State  Avail  JCTs Type       SigType TunnelName
10.123.123.1     100    Up     100%   10   MPLS-Sig   Compute R1toR9-cmp
10.123.123.1     101    Up     100%   8    MPLS-Sig   Compute R1toR9-margin-0-cmp
10.123.123.1     102    Up     100%   10   MPLS-Sig   RSVP    R1toR9
10.123.123.1     103    Up     100%   10   MPLS-Sig   RSVP    R1toR9-BW400m
10.123.123.1     104    Up     100%   9    MPLS-Sig   RSVP    R1toR9-BW400m-margin-10
10.123.123.1     105    Up     100%   8    MPLS-Sig   RSVP    R1toR9-margin-0
10.123.123.1     106    Up     100%   8    MPLS-Sig   RSVP    R1toR9-excl-dgo
10.123.123.1     107    Up     100%   7    MPLS-Sig   RSVP    R1toR9-incl-jgo
10.123.123.1     108    Up     100%   10   MPLS-Sig   RSVP    R1toR8
10.123.123.1     109    Up     100%   8    MPLS-Sig   RSVP    R1toR8-BW500m-excl-r7-r10

Total: 10
Up 10, Degraded 0, Down 0
```

About 1300 Mbps is now reserved on the links of R1.